

LECTURE 1 (SEPTEMBER 2ND)

Definition. (Contour integral) Let C be a contour parametrised by $z = z(t)$ where $a \leq t \leq b$. Also, let f be continuous on C . Then, the contour integral of f is defined as

$$\int_C f(z) dz = \int_C u dx - v dy + i \int_C u dy + v dx$$

and also, through parametrisation, we have

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

Definition. (Derivative) The derivative of a complex function is defined as

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

if it exists. If f is C^1 (the function is continuous up its first derivatives) and $u_x = v_y$ & $v_x = -u_y$, then $f'(z)$ exists.

Theorem. (Green's theorem) Let P and Q be a C^1 function on $\bar{D}$ where D is a simply connected bounded region whose boundary is C . Then,

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

where C is oriented in the positive sense. Here,

$$\int_C P dx = - \iint_D \frac{\partial Q}{\partial y} dy dx$$

and

$$\int_C Q dy = \iint_D \frac{\partial Q}{\partial x} dx dy$$

Proof. Notice that for a parametrised simple disc where the above is $y = g_2(x)$ and $y = g_1(x)$ on the bottom,

$$\int_C P dx = \int_a^b P(x, g_1(x)) dx - \int_a^b P(x, g_2(x)) dx$$

and

$$\iint_D \frac{\partial P}{\partial y} dy dx = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y} dy dx = \int_a^b \left[P(x, g_2(x)) - P(x, g_1(x)) \right] dx$$

Similarly,

$$\int_D Q \, dy = \int_c^d Q(y, h_2(x)) \, dy - \int_c^d Q(h_1(y), y) \, dy$$

and

$$\iint \frac{\partial Q}{\partial x} \, dx \, dy = \int_c^d \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x} \, dx \, dy = \left[Q(h_2(y), y) - Q(h_1(y), y) \right] dy$$

□

Theorem. (Cauchy's theorem) If f is analytic (differentiable) inside and on a bounded simply connected domain D whose boundary is C and $f \in C^1(\bar{D})$ then

$$\int_C f(z) \, dz = 0$$

Proof. Let $f = u + iv$. Then $u_x = v_y$ and $v_x = -u_y$ on D . Thus

$$\int_C f(z) \, dz = \int_C u \, dx - v \, dy + i \left[\int_C u \, dy + v \, dx \right]$$

where

$$\int_C u \, dx - v \, dy = \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

and

$$\int_C u \, dy + v \, dx = \iint_D \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

notice how the terms in the large brackets vanish, and that the integral itself equals zero. □

Theorem. (Generalised Cauchy's Theorem) If $f \in C^1(\bar{D})$ is analytic in a bounded multiply connected domain D with finitely many holes inside, then

$$\int_{\partial D} f(z) \, dz = 0$$

where ∂D is a positively oriented boundary of D .

Theorem. (Cauchy Integral Formula) If f is analytic inside and on a POSCC C then for every z_0 inside C we have

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} \, dz$$

LECTURE 2 (SEPTEMBER 4TH)

Recall. (Cauchy's formula) For a positively oriented simple closed curve (POSCC) C and f that is analytic inside and on C , we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{\xi - z} d\xi$$

for all z inside C . If $\bar{D}(z_0, r) = \{z \mid |z - z_0| \leq r\} \subset D$,

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0 + z_0 - z} = \frac{1}{(\xi - z_0) \left(1 - \frac{z - z_0}{\xi - z_0}\right)}$$

for $z \in D(z_0, r)$. However,

$$\left| \frac{z - z_0}{\xi - z_0} \right| < \frac{|z - z_0|}{r} < 1$$

so that

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}$$

Thus

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_C f(\xi) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}} d\xi \\ &= \sum_{n=0}^{\infty} a_n (z - z_0)^n \end{aligned}$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi = \frac{1}{n!} f^{(n)}(z_0)$$

Notice that the convergence is uniform for $\xi \in C$.

Theorem. If f_n is continuous in C such that $f_n \rightarrow f$ uniformly on C

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz$$

Theorem. If $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$ for $z \in D(z_0, R)$ then for $0 < r < R$ we have

$$\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt = \sum_{n=0}^{\infty} |c_n|^2 r^{2n}$$

Proof. Take

$$f(z_0 + re^{it}) = \sum_{n=0}^{\infty} c_n r^n e^{int}$$

so that

$$\begin{aligned}
\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} c_n r^n e^{int} \sum_{m=0}^{\infty} \bar{c}_m r^m e^{-imt} dt \\
&= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{1}{2\pi} \int_0^{2\pi} c_n \bar{c}_m r^{n+m} e^{i(n-m)t} dt \\
&= \sum_{n=0}^{\infty} |c_n|^2 r^{2n}
\end{aligned}$$

Here, we note that

$$\int_0^{2\pi} e^{i(n-m)t} dt = \begin{cases} 2\pi & n = m \\ 0 & n \neq m \end{cases}$$

□

Corollary. (Liouville theorem) All bounded entire functions are constant. Using this theorem, we can prove the fundamental theorem of algebra.

Proof. If f is entire such that $|f(z)| \leq M$ for all $z \in \mathbf{C}$ then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq M^2$$

for all $r > 0$ which means that $c_1 = c_2 = \dots = 0$.

□

Theorem. (Identity theorem) Let D be a domain (open and connected) and $\{z_n\}$ be a sequence of distinct points in D such that $z_n \rightarrow z_0 \in D$. If f is analytic in D with $f(z_n) = 0$ for all $n \in \mathbf{N}$, then $f(z) = 0$ for all $z \in D$.

Theorem. (Maximum modulus theorem) Let f be analytic in a domain D . Then, $|f|$ can not have a local maximum in D unless it is constant.

Proof. If

$$|f(z_0 + re^{it})| \leq |f(z_0)|$$

for all $t \in [0, 2\pi]$, then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq |f(0)|^2 = |c_0|^2$$

which means that $c_1 = \dots = c_n = 0$. Then we can apply the identity theorem.

□

Theorem. (Morera's theorem) Morera's theorem can be thought as Cauchy theorem's inverse. If f is continuous in a domain D such that

$$\int_C f(z) dz = 0$$

for all closed contours C in D , then f is analytic in D (note that this also means that the integral is path independent).

Proof. The key point is that for a given $a \in D$, $F(z) = \int_a^z f(\xi) d\xi$ is well defined. Take any $z_0 \in D$. We'll show that $F'(z_0) = f(z_0)$, which means that $f = F'$ is analytic. Take any $\varepsilon > 0$. We choose $\delta > 0$ so that

(i) $D(z_0, \delta) \subset D$

(ii) If $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \varepsilon$

We need to prove that if $0 < |z - z_0| < \delta$ then

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \varepsilon$$

Take the line segment path $[z_0, z]$ from z_0 to z . Then

$$F(z) - F(z_0) = \int_{[z_0, z]} f(\xi) d\xi$$

and

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \int_{[z_0, z]} [f(\xi) - f(z_0)] d\xi$$

Notice that now, due to (ii), the term inside the square brackets can be reduced to less than ε since $|\xi - z_0| < \delta$. Therefore

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \frac{1}{|z - z_0|} |z - z_0| \varepsilon = \varepsilon$$

□

Theorem. Let $\{f_n\}$ be a sequence of analytic functions in a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D . Then f is analytic on D .

Example. On $\mathbf{R}$, consider

$$f_n(x) = \sum_{k=1}^n \frac{\cos 3^k x}{2^k} \rightarrow f(x) = \sum_{n=1}^{\infty} \frac{\cos 3^n x}{2^n}$$

where the convergence is uniform and the right-hand side is nowhere differentiable.

Proof. First, f is continuous on D since convergence is uniform. Let C be any closed contour in D . Then

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz = 0$$

By Morera's theorem, the proof is complete. $\square$

LECTURE 3 (SEPTEMBER 9TH)

Lemma. Let f be analytic on a simply connected domain D such that $f(z) \neq 0$ for any $z \in D$. Then,

- (i) There is g analytic on D such that $e^g = f$ on D
- (ii) There is h analytic on D such that $h^2 = f$ on D (this implies the Riemann mapping theorem)

Proof. (1) By Cauchy's theorem, the $\int_C f(z) dz = 0$ for every closed contour C in D . Thus for a given $z_0 \in D$, the integral

$$\int_{z_0}^z f(\xi) d\xi$$

is independent of path. Since f has no zero in D , f'/f is analytic on D . We define g on D by

$$g(z) = \int_{z_0}^z \frac{f'(\xi)}{f(\xi)} d\xi + \text{Log } f(z_0)$$

Then we'll show that $f(z) = e^{g(z)}$ on D . Observe that

$$\begin{aligned} (fe^{-g})' &= f'e^{-g} + f(e^{-g})' \\ &= f'e^{-g} + fe^{-g}\left(-\frac{f'}{f}\right) \\ &= f'e^{-g} - f'e^{-g} = 0 \end{aligned}$$

on D and $fe^{-g} = c$ a constant. If we define $h(z) = f(z)e^{-g(z)}$ then evaluating $h(z)$ at z_0 gives

$$\begin{aligned} h(z_0) &= f(z_0)e^{-g(z_0)} \\ &= f(z_0)e^{-\text{Log } f(z_0)} \\ &= f(z_0)[f(z_0)]^{-1} = 1 \end{aligned}$$

Therefore, $f(z) = e^{g(z)}$.

(2) By (1), there exists an analytic g such that $f = e^g$ on D . If we define $h = e^{g/2}$, then h is analytic on D with $h^2 = f$. $\square$

Theorem. (Exhaustion of D by compact subsets) Let $D \subset \mathbf{C}$ be open. Then there is a sequence of compact subsets K_n of D such that

(i)

$$K_1 \subset K_2 \subset \dots \subset K_n \subset D$$

(ii)

$$\bigcup_{n=1}^{\infty} K_n = D$$

(iii) If $K \subset D$ is compact then $K \subset K_n$ for some $n \in \mathbf{N}$

(iv) There is a sequence $\{\delta_n\}$ with $\delta_n > 0$ such that

$$\bar{D}(z, 2\delta_n) \subset K_{n+1}$$

for each $z \in K_n$

Proof. Define

$$K_n = \left\{ z \in \mathbf{C} \mid |z| \leq n \right\} \cap \left\{ z \in \mathbf{C} \mid d(z, D^c) \geq \frac{1}{n} \right\}$$

$$G_n = \left\{ |z| < n \right\} \cap \left\{ d(z, D^c) > \frac{1}{n} \right\}$$

and $G_n = D$ by the Archimedean property. □

LECTURE 4 (SEPTEMBER 16TH)

Theorem. Let D be a domain in $\mathbf{C}$. There is a sequence $\{K_n\}$ of compact subsets of D and $\{\delta_n\}$ ($\delta_n > 0$) such that

(i)

$$\bigcup_n K_n = D$$

(ii) If $K \subset D$ is compact then $K \subset K_n$ for some n

(iii) $\bar{D}(z, 2\delta_n) \subset K_{n+1}$ for every $z \in K_n$, $n = 1, 2, \dots$

Note that $z \notin F$ and F is closed. This implies that $d(z, F) = \min\{|z - w| \mid w \in F\} > 0$.

Proof. Put

$$K_n = \left\{ |z| \leq n \right\} \cap \left\{ d(z, D^c) \geq \frac{1}{n} \right\}$$

□

Example. Consider

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \rightarrow 0$$

which converges uniformly on $\mathbf{R}$. However,

$$f'_n(0) = \sqrt{n} \rightarrow \infty$$

as $n \rightarrow \infty$.

Theorem. If $\{f_n\}$ is a sequence of analytic functions on a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D , then

- (i) f is analytic on D (by Morera's theorem)
- (ii) $f'_n \rightarrow f'$ uniformly on every compact subset of D

Proof. Take any compact subset K of D . Then $K \subset K_n$ for some n , and $\bar{D}(z, \delta_n) \subset K_{n+1}$ for all $z \in K$. So, there exists $r > 0$ and K' , a compact subset of D , such that $\bar{D}(z, r) \subset K'$ for all $z \in K$. Here, applying Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{|\xi-z|=r} \frac{f(\xi)}{(\xi-z)^2} d\xi$$

For every $z \in K$, we have

$$f'(z) = \frac{1}{2\pi i} \int_{|\xi-z|=r} \frac{f(\xi)}{(\xi-z)^2} d\xi$$

which implies that

$$|f'(z)| \leq \frac{1}{2\pi} \frac{\|f\|_{K'}}{r^2} (2\pi r) = \frac{\|f\|_{K'}}{r}$$

with $\|f\|_K = \max_{z \in K} |f(z)|$. Therefore, for every $z \in K$,

$$|f'_n(z) - f'(z)| \leq \frac{\|f_n - f\|_{K'}}{r}$$

which means $f'_n \rightarrow f'$ uniformly on K . □

Example. (Linear mapping) Consider the linear mapping

$$w = az + b$$

This mapping we see that, by expressing $a = re^{i\theta}$, the map stretches, rotates, and translates the complex plane.

Definition. (Linear functional transformation) For $ad - bc \neq 0$,

$$w = T(z) = \frac{az + b}{cz + d}$$

This can be written as

$$w = T(z) = \frac{a}{c} + \frac{bc - ad}{c} \cdot \frac{1}{cz + d}$$

From this, we understand that the map is 1-1 as the following inverse map exists:

$$z = \frac{dw - b}{-cw + a}$$

Additionally, we see that the mapping can be seen as a combination of translation, dilation ($|a| > 1$) or contraction ($|a| \leq 1$), rotation, and inversion.

$$z \rightarrow \frac{1}{z}$$

This is the tricky one!

Remark. (Inversion map $w = 1/z$) The equation (locus) of all lines and all circles in $\mathbf{R}^2$ is

$$a(x^2 + y^2) + bx + cy + d = 0$$

where $b^2 + c^2 > 4ad$ and $a, b, c, d \in \mathbf{R}$. In the complex plane, for $x^2 + y^2 = |z|^2$, $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/2i$ so that

$$a|z|^2 + \frac{1}{2}(b - ic)z + \frac{1}{2}(b + ic)\bar{z} + d = 0$$

and again $b^2 + c^2 > 4ad$. This implies that

$$a|z|^2 + \beta z + \bar{\beta}\bar{z} + d = 0$$

where $a, d \in \mathbf{R}$ and $\beta \in \mathbf{C}$ with $|\beta|^2 > ad$. Taking the inversion map,

$$a \frac{1}{|w|^2} + \beta \frac{1}{w} + \bar{\beta} \frac{1}{\bar{w}} + d|w|^2 = 0$$

and

$$a + \beta\bar{w} + \bar{\beta}w + d = 0$$

by multiplying $|w|^2 = w\bar{w}$ ($\beta\bar{\beta} > ad$). We therefore see that the inversion map takes the set of all lines and circles to again the set of all lines and circles.

Definition. (S^2) S^2 is defined as the boundary of a unit ball in $\mathbf{R}^3$. We see this as a one-point compactification (Alexandroff compactification) of $\mathbf{C}$, and is seen as $\mathbf{C} \cup \{\infty\}$ where $\{\infty\}$ is called the north pole.

Lemma. The linear functional transformation which takes $\{a, b, c\}$ to $\{0, 1, \infty\}$ is uniquely determined by

$$T(z) = \frac{(z-a)(b-c)}{(z-c)(b-a)}$$

LECTURE 5 (SEPTEMBER 18TH)

Definition. Notice that the linear fractional transformation defined for $ad - bc \neq 0$ as

$$T(z) = \frac{az + b}{cz + d}$$

we see that the derivative is

$$T'(z) = \frac{ad - bc}{(cz + d)^2} \neq 0$$

and that it is analytic and T' is never 0. Meanwhile,

$$T^{-1}(z) = \frac{dz - b}{-cz + a}$$

Conformal are by definition, angle preserving maps (both size and orientation).

Definition. Let $z = z(t)$ for $a \leq t \leq b$ represent the contour C . Also, let $w = f(z)$, that is, $w(t) = f(z(t))$ represent the contour Γ . If $z_0 = z(t_0)$ and $w_0 = f(z_0) = f(z(t_0))$, f is analytic at z_0 such that $f'(z_0) \neq 0$. Then, from

$$w'(t_0) = f'(z_0)z'(t_0)$$

we have

$$\arg w'(t_0) = \arg f'(z_0) + \arg z'(t_0)$$

Thus, we see that analytic functions with nonzero derivatives are conformal mappings.

Definition. Consider

$$T(z) = \frac{az + b}{cz + d}$$

(with $ad - bc \neq 0$) and

$$S(z) = \frac{ez + f}{gz + h}$$

(with $eh - gf \neq 0$) we ultimately have

$$(T \circ S)(z) = \frac{(ae + bg)z + (af + bh)}{(ce + dg)z + (cf + dh)}$$

also,

$$T^{-1}(z) = \frac{dz - b}{-cz + a}$$

Ultimately, we see that these mappings are equivalent to matrix multiplication.

Proposition. (The fixed points of $T(z) = z$) We see that

$$\frac{az + b}{cz + d} = z \quad \implies \quad cz^2 + (d - a)z - b = 0$$

and that there can't be more than two fixed points. If S and T are linear fractional transformations such that $S(z_k) = T(z_k)$ for $k = 1, 2, 3$, then $S = T$ since $S^{-1} \circ T = I$.

Theorem. If distinct points $\{z_1, z_2, z_3\}$ and distinct points $\{w_1, w_2, w_3\}$ are points of $S^2 = \mathbf{C} \cup \{\infty\}$ then there exists a unique linear fractional transformation T such that $T(z_k) = w_k$ for $k = 1, 2, 3$.

$$\{a, b, c\} \rightarrow \{0, 1, \infty\} \leftrightarrow \{\alpha, \beta, \gamma\}$$

Then

$$T(z) = \frac{(b - c)(z - a)}{(b - a)(z - c)} \quad \& \quad S(z) = \frac{(\beta - \gamma)(z - \alpha)}{(\beta - \alpha)(z - \gamma)}$$

Example. Find V such that

$$\{0, 1, 2\} \rightarrow \{-1, 0, 4\}$$

Proof.

$$S(z) = \frac{-4(z + 1)}{1(z - 4)} = \frac{-4z - 4}{z - 4}$$

(we say it maps to $\{0, 1, \infty\}$) which is equivalent to

$$\begin{pmatrix} -4 & -4 \\ 1 & -4 \end{pmatrix}$$

For T that sends $\{0, 1, 2\}$ to the same points, we find the matrix, and $V(z)$ is given as

$$V(z) = \frac{8z - 8}{-3z + 8}$$

□

Example. Let

$$T(z) = \frac{1 + z}{1 - z}$$

which corresponds to

$$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

Let $U = \{|z| < 1\}$, and say that we want to find $T(U)$. Notice that

$$\{-1, 0, 1\} \mapsto \{0, 1, \infty\}$$

As $T([-1, 1])$ maps to the positive x -axis, we can see that $T(-1) = \infty$ and that $T(|z| = 1)$ is a line passing $T(-1) = 0$. Then, $T(i) = i$ and $T(|z| = 1)$ is the y -axis. Another way to do this is through conformality of $T(|z| = 1)$ and the positive real axis being $\pi/2$ at $z = 0$.

Definition. Let U be the unit disk and $T = \{|z| = 1\}$ be a unit circle. For $\alpha \in U$, we define

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z} : \begin{pmatrix} 1 & -\alpha \\ -\bar{\alpha} & 1 \end{pmatrix}$$

Then ϕ_α is analytic on $\bar{U}$ such that $\phi_\alpha^{-1} = \phi_{-\alpha}$, $\phi_\alpha(0) = -\alpha$, $\phi_\alpha(\alpha) = 0$, $\phi_\alpha(U) = U$, $\phi_\alpha(T) = T$, and $\phi'_\alpha(0) = 1 - |\alpha|^2$, $\phi'_\alpha(\alpha) = 1/(1 - |\alpha|^2)$.

Proof. We'll show that $\phi_\alpha(T) = T$.

$$\begin{aligned} \phi_\alpha(e^{it}) &= \frac{e^{it} - \alpha}{1 - e^{-it}\bar{\alpha}} = \frac{e^{it}(1 - e^{-it}\alpha)}{1 - e^{-it}\bar{\alpha}} \\ |\phi_\alpha(e^{it})| &= 1 \end{aligned}$$

□

LECTURE 6 (SEPTEMBER 23RD)

Recall. Last class, we've took $U = \{|z| < 1\}$ and $T = \{|z| = 1\}$ with $\alpha \in U$. The map $\phi_\alpha(z)$ was defined as

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z} : \begin{pmatrix} 1 & -\alpha \\ -\bar{\alpha} & 1 \end{pmatrix}$$

We've noted that $1 - \bar{\alpha}z = 0$ implied that $z = \overline{(1/\alpha)} \notin \bar{U}$. You see that ϕ_α is analytic and one-to-one. We find explicitly that

$$\phi_\alpha^{-1} = \begin{pmatrix} 1 & \alpha \\ \bar{\alpha} & 1 \end{pmatrix}$$

such that $\phi_\alpha^{-1} = \phi_{-\alpha}$

$$\phi_\alpha^{-1}(z) = \phi_{-\alpha} = \frac{z + \alpha}{1 + \bar{\alpha}z}$$

and

$$\phi'_\alpha(z) = \frac{1 - |\alpha|^2}{(1 - \bar{\alpha}z)^2}$$

so that $\phi'_\alpha(0) = 1 - |\alpha|^2$ and $\phi'_\alpha(\alpha) = 1/(1 - |\alpha|^2)$. What about mappings from T ? If $z = e^{it}$, then

$$\phi_\alpha(e^{it}) = \frac{e^{it} - \alpha}{1 - \bar{\alpha}e^{it}} = \frac{e^{it} - \alpha}{e^{it}(e^{-it} - \bar{\alpha})} = \frac{1}{e^{it}} \frac{e^{it} - \alpha}{(e^{it} - \alpha)}$$

such that $|\phi_\alpha(e^{it})| = 1$ and $\phi_\alpha(T) \subset T$. Furthermore, we know that $\phi_{-\alpha}(T) \subset T$. We thus find, that $\phi_\alpha(T) = T$. With this information, via the maximum modulus theorem, we find that

$$\max_{z \in \bar{U}} |\phi_\alpha(z)| = \max_{z \in T} |\phi_\alpha(z)| = 1$$

which implies that

$$|\phi_\alpha(z)| < 1$$

for all $z \in U$.

Remark. We summarize like the following:

(i) For $\alpha \in U$,

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z}$$

(ii) $\phi_\alpha : U \rightarrow U$ is one-to-one, analytic and onto.

(iii) $\phi_\alpha(T) = T$

(iv) $\phi'_\alpha(0) = 1 - |\alpha|^2$

(v) $\phi'_\alpha(\alpha) = 1/(1 - |\alpha|^2)$

Theorem. (Schwartz lemma) Let $f : U \rightarrow U$ is analytic with $f(0) = 0$. For every $z \in U$, we have $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$. Furthermore, if one of the equalities hold, for some $z \in U \setminus \{0\}$ then f is a rotation.

$$f(z) = e^{i\theta} z$$

for some θ .

Proof. Notice that there exists a g analytic on U such that $f(z) = zg(z)$. For every fixed $z \in U$,

$$\begin{aligned} |g(z)| &= \left| \frac{f(z)}{z} \right| \leq \max_{\theta} \left| \frac{f(re^{i\theta})}{r} \right| \\ &\leq \frac{1}{r} \end{aligned}$$

if $|z| < r < 1$. Taking $r \rightarrow 1^-$, $|f(z)/z| \leq 1$. That is, $|f(z)| \leq |z|$ for all $z \in U$. Also, $f'(z) = g(z) + zg'(z)$ which implies that $f'(0) = g(0)$. This again implies that $|f'(0)| = |g(0)| \leq 1$. Notice how if the first equality holds, $|f(z)| = |z|$, and the function f only rotates the input. If the second equality holds, $|f'(0)| \leq 1$, and $|f(z)| \leq |z|$, meaning the same thing!

Notice the following. If $f(0) = \alpha$, then define $g(z) = \phi_\alpha(f(z))$. Then, $g : U \rightarrow U$ is analytic with $g(0) = 0$. Applying the Schwartz lemma to g , $|g'(0)| \leq 1$ implies that

$$|f'(0)| \leq 1 - |\alpha|^2$$

But,

$$g'(0) = \phi'_\alpha(f(0))f'(0) = \phi'_\alpha(\alpha)f'(0) = \frac{1}{1 - |\alpha|^2}f'(0)$$

□

Theorem. (Pick theorem) If $f : U \rightarrow U$ is analytic with $f(\alpha) = \beta$, then

$$|f'(\alpha)| \leq \frac{1 - |\beta|^2}{1 - |\alpha|^2}$$

The equality holds when

$$f(z) = \phi_{-\beta}(e^{i\theta}\phi_\alpha(z))$$

Proof. We define g on U so that $g(0) = 0$. Indeed,

$$g = \phi_\beta \circ f \circ \phi_{-\alpha}$$

is analytic from U into U with $g(0) = 0$. Applying the Schwartz lemma to g , we have $|g'(0)| \leq 1$. But

$$g'(0) = \left(\phi'_\beta[f(\phi_\alpha(0))] \right) f'(\phi_\alpha(0)) \phi'_\alpha(0) = \phi'_\beta(\beta) \cdot f'(0) \cdot \phi'_{-\alpha}(0) = \frac{1 - |\alpha|^2}{1 - |\beta|^2} f'(0) \leq 1$$

When does the equality hold? Let $g(z) = \lambda z$ for some $|\lambda| = 1$. Then

$$\phi_\beta(f(\phi_{-\alpha}(z))) = \lambda z$$

By putting $z \rightarrow \phi_\alpha(z)$,

$$\phi_\beta(f(z)) = \lambda \phi_\alpha(z)$$

and

$$\phi_{-\beta}[\phi_\beta(f(z))] = \phi_{-\beta}(\lambda \phi_\alpha(z))$$

and ultimately

$$f(z) = \phi_{-\beta}(\lambda \phi_\alpha(z))$$

□

Theorem. If f is one-to-one analytic from U onto U , then there exists $\alpha \in U$ and $\theta \in \mathbf{R}$ such that $f(z) = e^{i\theta}\phi_\alpha(z)$ for all $z \in U$.

LECTURE 7 (SEPTEMBER 25TH)

Theorem. (Schwartz-Pick theorem) Let $f : U \rightarrow U$ be an analytic function such that $f(\alpha) = \beta$. Then,

$$|f'(\alpha)| \leq \frac{1 - |\beta|^2}{1 - |\alpha|^2}$$

and the equality holds when

$$f(z) = \phi_{-\beta}(\lambda \phi_{\alpha}(z))$$

for some $\lambda \in T$.

Example. Observe that $f(x) = x^3$ is one-to-one on $\mathbf{R}$, but $f'(0) = 0$. Therefore, we see that on $\mathbf{R}$, f being one-to-one on an interval I doesn't necessarily imply that $f'(x) \neq 0$ for any $x \in I$. However, the converse works as if $f'(x) \neq 0$ for any $x \in I$, then f is one-to-one by the mean value theorem.

However, on $\mathbf{C}$, $f(z) = e^z$ satisfies $f'(z) \neq 0$ for any $z \in \mathbf{C}$ but f is not one-to-one. We claim that if f is analytic and one-to-one on a domain D , then $f'(z) \neq 0$ for any $z \in D$.

Theorem. If $f : U \rightarrow U$ is one-to-one analytic, meaning that $f'(z) \neq 0$ for any $z \in U$, then $f(z) = e^{i\theta} \phi_{\alpha}(z)$ for some $\alpha \in U$, $\theta \in \mathbf{R}$

Proof. Let $f(\alpha) = 0$. First define $g = f^{-1}$ on U . Then $g(f(z)) = z$ for all $z \in U$ and g is analytic on U . Due to the fact that the derivative is never zero, we can use the inverse function theorem, where we are able to write

$$\frac{g(w) - g(w_0)}{w - w_0} = \frac{z - z_0}{f(z) - f(z_0)} = \frac{1}{\frac{f(z) - f(z_0)}{z - z_0}}$$

and

$$g'(w_0) = \frac{1}{f'(z_0)}$$

since $f'(z_0) \neq 0$. We see that

$$g'(0)f'(\alpha) = 1$$

by the chain rule. By the Schwartz-Pick theorem,

$$|f'(\alpha)| \leq \frac{1}{1 - |\alpha|^2}$$

and $|g'(0)| \leq 1 - |\alpha|^2$. This means that

$$\frac{1}{1 - |\alpha|^2} \leq |f'(\alpha)| \leq \frac{1}{1 - |\alpha|^2}$$

By the Schwartz lemma,

$$f(z) = \lambda \phi_\alpha(z)$$

for some $\lambda \in T$. □

Theorem. If f is analytic in a domain D with $f'(z_0) \neq 0$ for some $z_0 \in D$ then there is $r > 0$ such that $D(z_0, r) \subset D$ and f is one-to-one on $D(z_0, r)$.

Proof. If $z, w \in D(z_0, r)$ and $z \neq w$, then

$$\left| \frac{f(z) - f(w)}{z - w} - f'(z_0) \right| < \frac{1}{2} |f'(z_0)|$$

which implies that

$$\frac{1}{2} |f'(z_0)| < \left| \frac{f(z) - f(w)}{z - w} \right|$$

if $z, w \in D(z_0, r)$ with $z \neq w$. □

Lemma. If f is analytic in a domain D and

$$g(z, w) = \begin{cases} \frac{f(z) - f(w)}{z - w} & z \neq w \\ f'(z) & z = w \end{cases}$$

then g is continuous on $D \times D$. Note that this is trivially true on $\mathbf{R}$.

Lemma. If $|b - a| < \frac{|a|}{2}$, then $|b| > \frac{|a|}{2}$.

Proof. Notice that it suffices to prove the statement only when $z = w$. Since f' is continuous on D , for every ε , there exists a δ such that if $z \in D(a, \delta)$ then $|f'(z) - f'(a)| < \varepsilon$. Take any z, w ($z \neq w$) in $D(a, r)$. Define $\xi(t) = tz + (1 - t)w$ for $0 \leq t \leq 1$. Then,

$$\int_0^1 f'(\xi(t)) dt = \frac{f(z) - f(w)}{z - w} = g(z, w)$$

so that

$$g(z, w) - g(a, a) = \int_0^1 [f'(\xi(t)) - f'(a)] dt$$

Since $z, w \in D(a, r)$ we have

$$\left| \int_0^1 [f'(\xi(t)) - f'(a)] dt \right| \leq \int_0^1 |f'(\xi(t)) - f'(a)| dt < \varepsilon$$

In conclusion, if $z, w \in D(a, r)$, then $|g(z, w) - g(a, a)| < \varepsilon$. □

Theorem. (Rouche's theorem) If f and g are analytic inside and on a simple, closed contour C such that $|g(z)| < |f(z)|$ for every $z \in C$, then f and $f + g$ have the same number of zeros inside C .

Theorem. (Open mapping theorem) Let f be non-constant analytic in a domain Ω . If $D \subset \Omega$ is open, then $f(D)$ is open.

Proof. To show that $f(D)$ is open, take any $w_0 \in f(D)$. We need to find some $m > 0$ such that $D(w_0, m) \subset f(D)$. Let $f(z_0) = w_0$. Then there is $r > 0$ such that

$$\bar{D}(z_0, r) \subset D$$

and $f(z) - w_0$ has no zeros in $0 < |z - z_0| \leq r$ by the identity theorem. Let

$$m = \min_{|z - z_0| = r} |f(z) - w_0| > 0$$

Then we'll show that $D(w_0, m) \subset f(D)$. Take any $w_1 \in D(w_0, m)$ (we're trying to show that $w_1 \in f(D)$). Then

$$|w_0 - w_1| < |f(z) - w_0|$$

on $|z - z_0| = r$. Rouché's theorem implies that $f(z) - w_0$ and $f(z) - w_1$ have the same number of zeros inside $|z - z_0| = r$ which implies that

$$w_1 \in f(D(z_0, r)) \subset f(D)$$

□

Theorem. If f is one-to-one analytic in a domain D , then $f'(z) \neq 0$ for any $z \in D$.

Proof. Suppose that $f'(z_0) = 0$ for some $z_0 \in D$ (we'll show that f is not one-to-one). Then $f(z) - f(z_0) = f''(z_0)(z - z_0)^2 + \dots$ in some neighborhood of z_0 . There is $r > 0$ such that $\bar{D}(z_0, r) \subset D$ and $f(z) - f(z_0)$ and $f'(z)$ both have no zeros in $0 < |z - z_0| \leq r$.

Let $m = \min_{|z - z_0| = r} |f(z) - f(z_0)|$ and choose any $w_1 \in D(f(z_0), m)$. Then,

$$|f(z_0) - w_1| < m \leq |f(z) - f(z_0)|$$

on $|z - z_0| = r$. By Rouché's theorem, $f(z) - f(z_0)$ and $f(z) - w_1$ have the same number of zeros in $D(z_0, r)$ where $f(z) - f(z_0)$ has at least 2 zeros. In other words, since $f' \neq 0$ in $D(z_0, r)$, there is at least 2 distinct zeros of $f(z) - w_1$ in $D(z_0, r)$ where f is not one-to-one. □

LECTURE 8 (SEPTEMBER 30TH)

Theorem. Let $\{f_n\}$ be a sequence of 1-1 analytic functions on a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D . Then, either f is constant or f is 1-1 on D .

Proof. Let f be non-constant. Suppose that there exists $z_1, z_2 \in D$ such that $f(z_1) = f(z_2) = a$ and that the function is not 1-1. Then, there exists $r > 0$ such that:

(i) $\bar{D}(z_1, r) \subset D$ and $\bar{D}(z_2, r) \subset D$

(ii) By the identity theorem, $f(z) - a$ has no zeros in both $0 < |z - z_1| \leq r$ and $0 < |z - z_2| \leq r$. Denote simple circles $C_1 : |z - z_1| = r$ and $C_2 : |z - z_2| = r$. Let

$$m = \min_{z \in C_1 \cup C_2} |f(z) - a| > 0$$

where $C_1 \cup C_2$ is compact.

(iii) $\bar{D}(z_1, r) \cap \bar{D}(z_2, r) = \emptyset$

By the uniform convergence of $\{f_n\}$ on $C_1 \cup C_2$, there is N such that

$$|f_N(z) - f(z)| < m$$

for $z \in C_1 \cup C_2$. On C_1 , $|f_N(z) - f(z)| < m \leq |f(z) - a|$. By Rouché's theorem, $f(z) - a$ and $f_N(z) - a$ have the same numbers of zero inside C_1 . Also, $f(z) - a$ and $f_N(z) - a$ have the same number of zeros inside C_2 . Thus, $f_N(z) - a$ has zeros inside C_1 and inside C_2 . Thus, f_N is not 1-1. $\square$

Example. There are multiple cases where the first option holds. For example, take $f_n(z) = \frac{z}{n}$ and $f = 0$.

Theorem. (Riemann mapping theorem) If $D (\neq \mathbf{C})$ is a simply connected domain, then there is a 1-1 analytic function h from D onto U .

Proof. Let $\Sigma = \{f : D \rightarrow U \mid f \text{ is injective and analytic}\}$.

(I) We prove that $\Sigma \neq \emptyset$. To prove this, we use the prior result that if f is analytic on a simply connected domain D , such that $f(z) \neq 0$ for any $z \in D$, then there is g analytic in D such that $g^2 = f$ on D . Suppose that $w_0 \notin D$. Then there is g analytic on D such that $g^2(z) = z - w_0$. As $w^2 = z$ implies that $w = \sqrt{r}e^{i\theta/2}$, $g(D)$ does not intersect some half of the plane! Furthermore,

(i) $0 \notin g(D)$, which is trivial.

- (ii) g is 1-1 on D , as if $g(z_1) = g(z_2)$, then $g^2(z_1) = g^2(z_2)$ and $z_1 - w_0 = z_2 - w_0$.
- (iii) If $w \in g(D)$, then $-w \notin g(D)$, as if $g(z_1) = -g(z_2)$ for distinct points, then $g^2(z_1) = g(z_2)^2$ and $z_1 = z_2$ which is a contradiction.
- (iv) If $a \in g(D)$, then there exists an $r > 0$ such that $D(a, r) \subset g(D)$ due to the open mapping theorem. In particular, $D(-a, r) \cap g(D) = \emptyset$ as if $w \in D(-a, r)$, then $|w + a| < r$ and $|-w - a| = |w + a| < r$ which implies that $-w \in D(a, r) \subset g(D)$.

Now, define $h : D \rightarrow \mathbb{C}$ by

$$h(z) = \frac{r}{g(z) + a}$$

Notice that $|g(z) + a| > r$ for all $z \in D$ and that $|h(z)| < 1$ for all $z \in D$. We see that if $h(z_1) = h(z_2)$.

$$\frac{r}{g(z_1) + a} = \frac{r}{g(z_2) + a}$$

and that $g(z_1) = g(z_2)$ with $z_1 = z_2$. We see that $h(z)$ is a 1-1 function from an arbitrary domain D to U !

(II) Let $\psi \in \Sigma$ such that $\psi(D) \neq U$. We claim that for every $z_0 \in D$, there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$. Suppose that $\alpha \in U \setminus \psi(D)$. Then, $\phi_\alpha \circ \psi$ is analytic on D with no zeros. Since D is simply connected, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$. Let $g(z_0) = \beta$. Define $\psi_0 = \phi_\beta \circ g$. Then, $\psi_0 \in \Sigma$ and due to the Schwartz lemma, the claim holds. $\square$

LECTURE 9 (OCTOBER 2ND)

Proof. (I) Take any $w_0 \notin D$, and then there is g that is analytic in D such that $g^2(z) = z - w_0$. If $D(a, r) \subset g(D)$, then $D(-a, r) \cap g(D) = \emptyset$. Now define

$$h(z) = \frac{r}{g(z) + a}$$

then $h \in \Sigma$.

(II) We claim that if $\psi \in \Sigma$ and $\psi(D) \neq U$, then for every $z_0 \in D$, there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$. Let $\alpha \notin \psi(D)$, then $\phi_\alpha \circ \psi$ is analytic in D with no zeros in D . Thus, there is g analytic in D such that $g^2 = \phi_\alpha \circ \psi$ in D . Given $z_0 \in D$, if $\beta = g(z_0)$, then $\psi_0 = \phi_\beta \circ g$ satisfies the claim.

Notice how $g \in \Sigma$ since g is 1-1 in D (if $g(z_1) = g(z_2)$, then $(\phi_\alpha \circ \psi)(z_1) = (\phi_\alpha \circ \psi)(z_2)$ concluding that $z_1 = z_2$). Additionally denote the square function by $s(z) = z^2$. Then,

$$\phi_\alpha \circ \psi = g^2 = s \circ g$$

so that $\psi = \phi_{-\alpha} \circ s \circ g$. Additionally, $\psi_0 = \phi_{\beta} \circ g$ so that $g = \phi_{-\beta} \circ \psi_0$. Therefore,

$$\psi = \phi_{-\alpha} \circ s \circ \phi_{-\beta} \circ \psi_0$$

on D . Now we will show that the claimed inequality holds. Defining $F = \phi_{\alpha} \circ s \circ \phi_{-\beta}$ or $F : U \rightarrow U$ is analytic. By the Schwartz lemma,

$$|F'(0)| \leq 1$$

since F is not 1-1. From $\psi = F \circ \psi_0$, we have

$$\psi'(z_0) = F'(\psi_0(z_0))\psi_0'(z_0) = F'(0)\psi_0'(z_0)$$

Since ψ is 1-1, the derivative $\psi' \neq 0$ in D and we have proved the claim.

(III) We now claim that if we take any $z_0 \in D$ and let $\eta_0 = \sup\{|\psi'(z_0)| \mid \psi \in \Sigma\}$, there is $h \in \Sigma$ such that $|h'(z_0)| = \eta_0$. From this, as $h(D)$ doesn't conform to the condition of step (II), we will have that $h(D) = U$. The key method here is using "normal families". $\square$

Remark. On $\mathbf{R}^n$,

- (i) If G is a bounded infinite, then G has a limit point
- (ii) If $\{a_n\}$ is a bounded sequence, then $\{a_n\}$ has a convergent subsequence
- (iii) If K is closed and bounded, then every open cover of K has a finite subcover

However, consider the following. The Hilbert space $l^2(\mathbf{N})$ can be seen as the space closest to $\mathbf{R}^\infty$.

$$\mathbf{x} = (x_1, x_2, \dots) \in l^2(\mathbf{N}) \quad \text{if} \quad \|\mathbf{x}\|^2 = \sum_{n=1}^{\infty} x_n^2 < \infty$$

Defined this manner, $l^2(\mathbf{N})$ is a complete normed vector space. Consider $E = \{\mathbf{e}_n \mid n \in \mathbf{N}\} \subset l^2(\mathbf{N})$ where $\mathbf{e}_k = (0, \dots, 0, 1, 0, \dots)$ where only the k -th element is nonzero. This basis satisfies $\|\mathbf{e}_k\| = 1$.

- (i) Then, E is a bounded infinite subset of $l^2(\mathbf{N})$ which has no limit point.
- (ii) Also, E is a bounded sequence in $l^2(\mathbf{N})$ with no convergent subsequence.
- (iii) Lastly, E is closed and bounded in $l^2(\mathbf{N})$ but is not compact.

Definition. (Three types of convergences & boundednesses) If D is open and $f_n : D \rightarrow \mathbf{C}$, then we can define three types of convergences.

- (i) Pointwise convergence in D
- (ii) Uniform convergence in D

(iii) Uniform convergence in every compact subset of D

There are also three types of boundednesses of $\{f_n\}$ on D

(i) Uniformly bounded on D , meaning that $|f_n| \leq M$ for all $n \in \mathbf{N}$ and $z \in D$.

(ii) Pointwise bounded on D , meaning there is $M_z > 0$ such that $|f_n(z)| \leq M_z$ for every $n \in \mathbf{N}$.

(iii) Uniformly bounded on every compact subset of D , meaning that for every compact subset of K of D , there is $M_K > 0$ such that $|f(z)| \leq M_K$ for all $n \in \mathbf{N}$ and $z \in K$.

Theorem. (Arzela-Ascoli theorem) A separable metric space is space that has a countable dense subset and a metric space is a space endowed with a metric. If $\{f_n\}$ is a pointwise bounded and equicontinuous sequence of functions on a separable metric space, $\{f_n\}$ has a subsequence that converges uniformly on all compact subsets.

LECTURE 10 (OCTOBER 14TH)

Theorem. (Arzela Ascoli) Let $\{f_n\}$ be a pointwise bounded and equicontinuous sequence of functions on a separable metric space X . Then, $\{f_n\}$ has a subsequence which converges uniformly on all compact subsets of X . This subsequence converges pointwise since every $\{x_n\} \subset X$ is compact.

Definition. (Pointwise bounded) A sequence $\{f_n\}$ is pointwise bounded on X provided that for every $x \in X$ there is $M_x > 0$ such that $|f_n(x)| \leq M_x$ for all $n \in \mathbf{N}$.

Definition. (Equicontinuity) A sequence $\{f_n\}$ is equicontinuous on X provided that for every $\varepsilon > 0$, there is $\delta > 0$ such that if $d(x, y) < \delta$ then $|f_n(x) - f_n(y)| < \varepsilon$ for all $n \in \mathbf{N}$.

Remark. If $\{f_n\}$ is equicontinuous on X , then each f_n is uniformly continuous on X . The converse is not true, for example set $f_n : \mathbf{R} \rightarrow \mathbf{R}$ as

$$f_n(x) = nx$$

Notice how $|f_n(x) - f_n(y)| = n|x - y|$ and each f_n is uniformly continuous, but $\{f_n\}$ is not equicontinuous on $\mathbf{R}$.

Proof. Let $E = \{x_1, x_2, \dots\}$ be a countable dense subset of X . The proof is done in two steps:

(Step 1) We need to use pointwise boundedness to find a subsequence $\{g_n\}$ of $\{f_n\}$ which converges pointwise on E . Notice that $\{f_n(x_1)\}$ is a bounded sequence in $\mathbf{C}$, such that it has a convergent subsequence $\{f_{n,1}(x_1)\}$. Then, $\{f_{n,1}(x_2)\}$ is bounded and has a bounded

subsequence in $\mathbf{C}$, such that it has a convergent subsequence $\{f_{n,2}(x_2)\}$. Notice that meanwhile, $\{f_{n,2}(x_1)\}$ also converges. Continuing and taking the diagonal, we have $\{f_{n,n}(x)\}$ which converges for every $x_n \in E$. Set $g_n = f_{n,n}$, and we see that for any $x_m, g_m, g_{m+1}, \dots$ is a subsequence of $\{f_{n,m}\}_{n=1}^\infty$ which converges at x_m .

(Step 2) We now use equicontinuity to prove that this sequence converges uniformly on any compact subset K of X . We now use equicontinuity to show that there exists a $N \in \mathbf{N}$ satisfying if $n, m > N$ then $|f_n(x) - f_m(x)| < \varepsilon$ for every $x \in K$ ($\{f_n\}$ is uniformly Cauchy on K). Observe that due to the equicontinuity of $\{g_n\}$, for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $p, q \in X$ satisfies $d(p, q) < \delta$, then $|f_n(p) - f_n(q)| < \varepsilon/3$. Also,

$$\mathcal{F} = \left\{ B\left(x, \frac{\delta}{2}\right) \mid x \in K \right\}$$

is an open cover of K so that there is a finite subcover $\{B_1, B_2, \dots, B_M\}$ (here, note that if $x, y \in B_k$, then $d(x, y) < \delta$). Since E is dense in X , then $p_i \in E \cap B_i$ for $1 \leq i \leq M$. From step 1, $\lim_{n \rightarrow \infty} g_n(p_i)$ converges for $1 \leq i \leq M$ and thus $\{g_n(p_i)\}$ is a Cauchy sequence for $1 \leq i \leq M$. Thus there is $N \in \mathbf{N}$ such that if $n, m > N$ then $|g_n(p_i) - g_m(p_i)| < \varepsilon/3$ for $1 \leq i \leq M$ with $N = \max\{N_1, \dots, N_M\}$. Now we show the same for an arbitrary $x \in K$.

$$\begin{aligned} |g_n(x) - g_m(x)| &= |g_n(x) - g_n(p_k) + g_n(p_k) - g_m(p_k) + g_m(p_k) - g_m(x)| \\ &\leq |g_n(x) - g_n(p_k)| + |g_n(p_k) - g_m(p_k)| + |g_m(p_k) - g_m(x)| \\ &\leq \varepsilon \end{aligned}$$

as $d(x, p_k), d(x, p_m) < \delta$. □

Definition. (Uniformly bounded on a compact K) For every compact $K \subset D$, there is $M_K > 0$ such that $|f_n(z)| \leq M_K$ for all $z \in K$, $n \in \mathbf{N}$.

Theorem. (Montel's theorem) If $\{f_n\}$ is a sequence of analytic functions on a domain D which is uniformly bounded on each compact subset of D , then $\{f_n\}$ has a subsequence which converges uniformly on all compact subsets of D .

Theorem. If $\{f_n\}$ is a sequence of analytic functions on a domain D which is uniformly bounded on a compact subset of D , then $\{f_n\}$ is equicontinuous on every compact subset of D .

Proof. The proof is done through the Cauchy integral formula along with the exhaustion of D by compact subsets. □

LECTURE 11 (OCTOBER 16TH)

Theorem. Let $\{f_n\}$ be analytic in D uniformly bounded on each compact subset. Then, $\{f_n\}$ is equicontinuous on every compact subset of D .

Proof. Take any compact subset K of D . We'll show that $\{f_n\}$ is equicontinuous on K . Take any $\varepsilon > 0$, and we have to find $\delta > 0$ such that if $z, w \in K$ satisfies $|z - w| < \delta$, then $|f_n(z) - f_n(w)| < \varepsilon$ for all $n \in \mathbf{N}$.

By the exhaustion of compact subsets, there is a sequence of compact sets $\{K_n\}$ and $\delta_n > 0$ such that $\bigcup_{n=1}^{\infty} K_n = D$, and for every $z \in K_n$ we have $\bar{D}(z, 2\delta_n) \subset K_{n+1}$ for each n . Take $N \in \mathbf{N}$ such that $K \subset K_N$ so that for $z, w \in K$ and $|z - w| < \delta_N$ we have

$$f_n(z) = \frac{1}{2\pi i} \int_{|\xi-z|=2\delta_N} \frac{f_n(\xi)}{\xi-z} d\xi \quad \& \quad f_n(w) = \frac{1}{2\pi i} \int_{|\xi-w|=2\delta_N} \frac{f_n(\xi)}{\xi-w} d\xi$$

(where w is inside $|\xi - z| = 2\delta_N$) so that for $z, w \in K$ with $|z - w| < \delta_N$,

$$f_n(z) - f_n(w) = \frac{z-w}{2\pi i} \int_{|\xi-z|=2\delta_N} \frac{f_n(\xi)}{(\xi-z)(\xi-w)} d\xi$$

Noting that $|f_n(\xi)| \leq M_{K_{N+1}}$ and $|\xi - z| = 2\delta_N$ implying $|\xi - w| > \delta_N$, we have

$$|f_n(z) - f_n(w)| \leq \frac{|z-w|}{2\pi} \frac{M_{K_{N+1}}}{(2\delta_N)\delta_N} \cdot 4\pi\delta_N = \frac{M_{K_{N+1}}}{\delta_N} |z-w|$$

So far we have proved that if $z, w \in K$ satisfies $|z - w| < \delta_N$, then

$$|f_n(z) - f_n(w)| \leq \frac{M_{N+1}}{\delta_N} |z-w|$$

for all $n \in \mathbf{N}$. Take

$$\delta_n = \min \left\{ \delta_N, \frac{\delta_N \varepsilon}{M_{K_{N+1}}} \right\}$$

□

Theorem. (Montel's theorem) Let $\{f_n\}$ be analytic in a domain D , which is uniformly bounded on each compact subset of D . Then $\{f_n\}$ is a subsequence which converges uniformly on all compact subsets of D .

Proof. Use the Arzela Ascoli theorem and there is a subsequence $\{f_{n_k}\}$ which converges uniformly on K . □

Example. Let $D = \{\operatorname{Re}(z) > 0\}$, the right half plane. Let $f : D \rightarrow U$ be analytic with $f(1) = \alpha$. Find the upper bound of $|f'(1)|$.

Proof. Take

$$\phi = \frac{z-1}{z+1}$$

Using Schwartz's lemma for $F = \phi_\alpha \circ f \circ \phi^{-1}$, $F(0) = 0$ and

$$|F'(0)| \leq 1$$

with

$$F'(0) = \phi'_\alpha(\alpha) f'(1) \phi'(0) = \frac{1}{1-|\alpha|^2} f'(1) 2$$

Concluding,

$$|f'(1)| \leq \frac{1-|\alpha|^2}{2}$$

□

Example. Let Ω be the upper halfplane $\{z \mid \text{Im}(z) > 0\}$ and $f : \Omega \rightarrow U$ be an analytic function with $f(2i) = \alpha$. Find the upper bound of $|f'(2i)|$.

Proof. First find a linear fractional transformation ψ from Ω onto U satisfying $\psi(2i) = 0$. Consider

$$\psi(z) = \frac{z-2i}{z+2i}$$

For any $x \in \mathbf{R}$, $|\psi(x)| = 1$. If $\alpha \in \Omega$ ($\text{Im } \alpha > 0$), then

$$\psi(z) = \frac{z-\alpha}{z-\bar{\alpha}}$$

satisfies

$$|\psi(x)| = \left| \frac{x-\alpha}{x-\bar{\alpha}} \right| = 1$$

for $x \in \mathbf{R}$. From this we understand that ψ is a linear fractional transformation from $\{\text{Im } z > 0\}$ onto U . Thus, construct $F : U \rightarrow U$ that is analytic with $F(0) = 0$. We use

$$\psi = \frac{z-2i}{z+2i}$$

satisfying $\psi(\Omega) = U$, $\psi^{-1}(U) = \Omega$, and $\psi^{-1}(0) = 2i$.

$$F = \phi_\alpha \circ f \circ \psi^{-1}$$

which leads to $F(0) = 0$ and

$$F'(0) = \phi'_\alpha(\alpha) \cdot f'(2i) \cdot [\psi^{-1}]'(0)$$

Concluding,

$$|f'(2i)| \leq \frac{1-|\alpha|^2}{4}$$

□

LECTURE 12 (OCTOBER 28TH)

Remark. We continue with the final step of the Riemann mapping theorem. Let Σ be the set of all 1-1 analytic functions from D into U . We have proved if that $\phi \in \Sigma$ satisfies $\phi(D) \neq U$, then for every $z_0 \in D$ there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\phi'(z_0)|$.

(Step 3) If $z_0 \in D$ and $\eta_0 = \sup\{|\phi'(z_0)| \mid \phi \in \Sigma\}$, then there is $h \in \Sigma$ such that $|h'(z_0)| = \eta_0$, which implies that $h(D) = U$.

Proof. We suggest that $\mathcal{A} = \{|\phi'(z_0)| \mid \phi \in \Sigma\}$ is bounded. Suppose that it is not bounded. Then for every $n \in \mathbf{N}$, there is $\phi_n \in \Sigma$ such that $|\phi'_n(z_0)| > n$. However, notice that $|\phi_n(z)| < 1$ for all $z \in D$, $n \in \mathbf{N}$ and is uniformly bounded. By Montel's theorem, there is a subsequence $\{\phi_{n_k}\}$ which converges uniformly on every compact subset of D . Let $h(z) = \lim_{k \rightarrow \infty} \phi_{n_k}(z)$ for $z \in D$. Then, $h(z)$ is analytic in D and $\lim_{n \rightarrow \infty} \phi'_{n_k}(z) = h'(z)$ for every $z \in D$ so that $h'(z_0) = \lim_{k \rightarrow \infty} \phi'_{n_k}(z_0)$ which means that

$$|h'(z_0)| = \lim_{k \rightarrow \infty} |\phi'_{n_k}(z_0)| \geq \lim_{k \rightarrow \infty} n_k = \infty$$

which is a contradiction.

Knowing that $\eta_0 > 0$ (since $\phi' \neq 0$ for every $\phi \in \Sigma$) there is a subsequence $\{\phi_n\}$ in Σ such that $\lim_{n \rightarrow \infty} |\phi'_n(z_0)| = \eta_0$. Since $|\phi_n(z)| \leq 1$ for all $z \in D$, there is a subsequence $\{\phi_{n_k}\}$ which converges uniformly on all compact subsets of D to an analytic function h on D . Since $\lim_{k \rightarrow \infty} \phi'_{n_k}(z) = h'(z)$ for every $z \in D$, we have

$$|h(z_0)| = \lim_{k \rightarrow \infty} |\phi'_{n_k}(z_0)| = \eta_0$$

It remains to prove that $h \in \Sigma$. For every $z \in D$, $|h(z)| = \lim_{k \rightarrow \infty} |\phi_{n_k}(z)| \leq 1$ for all $z \in D$ as ϕ_{n_k} is a function from D to U . If $|h(z_1)| = 1$ for some $z_1 \in D$, by the maximum modulus theorem or the open mapping theorem, h is constant but $|h'(z_0)| = \eta_0 > 0$. This is again a contradiction. Thus, $h : D \rightarrow U$. Since ϕ_n is 1-1 analytic and $\phi_n \rightarrow h$ uniformly on all compact sets, we get that h is 1-1. □

Example. Assume that $h(z_0) = 0$. If $h(z_0) = \beta \neq 0$ then $f = \phi_\beta \circ h \in \Sigma$ satisfies

$$|f'(z_0)| = |\phi'_\beta(\beta)h'(z_0)| = \frac{h'(z_0)}{1 - |\beta|^2} > |h'(z_0)|$$

Example. Prove or disprove whether for a $f : D \rightarrow U$ is analytic when $|f'(z_0)| \leq |h'(z_0)|$.

Proof. This is true. Define $g = f \circ h^{-1}$. Then, $g : U \rightarrow U$ is analytic.

$$g'(0) = f'(z_0)(h^{-1})'(0) = f'(z_0) \frac{1}{h'(z_0)} = \frac{f'(z_0)}{h'(z_0)}$$

since $|g'(0)| \leq 1$ which implies that $|f'(z_0)| \leq |h'(z_0)|$. □

Theorem. If $g : U \rightarrow U$ is 1-1 onto analytic with $g(0) = 0$ and $g'(0) > 0$, then $g(z) = z$.

Theorem. If D is simply connected ($D \neq \mathbf{C}$) then for every $z_0 \in D$, there is a unique 1-1 analytic function h from D onto U satisfying $h(z_0) = 0$ and $h'(z_0) > 0$.

Proof. If $h \in \Sigma$ satisfies $|h'(z_0)| = \sup\{|\phi'(z_0)| \mid \phi \in \Sigma\}$, then $h(D) = U$ and $h(z_0) = 0$. If there exists f and g both satisfying the conditions above, define $h = f \circ g^{-1}$ and $h(0) = 0$ and $h'(0) = h'(z_0)/g'(z_0) > 0$ giving $h(z) = z$. □

LECTURE 13 (OCTOBER 30TH)

Theorem. Let $h'(z_0) = \sup\{|\phi'(z_0)| \mid \phi \in \Sigma\}$ then $h(\Omega) = U$ and $h \in \Sigma$. In addition, $h(z_0) = 0$. In addition, if we add the condition that $h'(z_0) > 0$, such h is uniquely determined.

Proof. If $h'(z_0) = re^{i\theta}$, then $f(z) = e^{-i\theta}h(z)$ satisfies $f \in \Sigma$, $f(\Omega) = U$, $f(z_0) = 0$ and $f'(z_0) = r > 0$. Let's prove uniqueness. If there exists f and g satisfying all conditions, then define

$$h(z) = f(g^{-1}(z))$$

and $h : U \rightarrow U$ is 1-1 analytic and onto.

$$h'(0) = \frac{f'(z_0)}{g'(z_0)} > 0$$

and $h(0) = 0$. This implies that $h(z) = 0$ and that $f = g$. □

Theorem. Let D be simply connected and $D \neq \mathbf{C}$. Then there is $\phi : D \leftarrow U$ that is 1-1 analytic and onto. Recall that if ϕ is 1-1, $\phi' \neq 0$ and is conformal. This tells us that there exists $\psi = \phi^{-1} : U \rightarrow D$ that is still 1-1 analytic and onto, thus conformal. Define $A(D)$ be the set of all analytic functions on D . Define $\Phi : A(D) \rightarrow A(U)$ by $\Phi(f) = f \circ \psi$. Then, Φ becomes a ring isomorphism.

Remark. Consider $\mathbf{C}^2 = \mathbf{C} \times \mathbf{C}$. The simplest domains in $\mathbf{C}^2$ are B_2 and U^2 .

$$B_2 = \{z = (z_1, z_2) \in \mathbf{C}^2 \mid \|z\| < 1\} \quad \& \quad U^2 = U \times U$$

each called a bidisc and a unit ball. The norm $\|z\| = \langle z, z \rangle^{1/2}$ is defined as

$$\langle z, z \rangle^{1/2} = \langle (z_1, z_2), (z_1, z_2) \rangle^{1/2} = \left(z_1 \bar{z}_1 + z_2 \bar{z}_2 \right)^{1/2} = \left(|z_1|^2 + |z_2|^2 \right)^{1/2}$$

Theorem. (Residue theorem) If f is analytic inside and on a POSCC C except for isolated singularities at $z_1, \dots, z_n$ inside C , then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}_{z=z_k} f(z)$$

If $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ in $0 < |z - z_k| < \delta$ (Laurent series of f at z_k), then

$$a_{-1} = \text{Res}_{z=z_k} f(z)$$

If $\lim_{z \rightarrow z_0} (z - z_0)^k f(z) = A \neq 0$, we say that f has a pole of order k at z_0 . If f has a simple pole (order 1) at z_0 , then $\text{Res}_{z=z_0} f(z) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$. If

$$f(z) = \frac{p(z)}{q(z)}$$

(p, q analytic at z_0) has a simple pole at z_0 ($p(z_0) \neq 0$, $q(z_0) = 0$, and $q'(z_0) \neq 0$), then

$$\text{Res}_{z=z_0} f(z) = \lim_{z \rightarrow z_0} (z - z_0) \frac{p(z)}{q(z)} = \lim_{z \rightarrow z_0} \frac{p(z)}{\frac{q(z) - q(z_0)}{z - z_0}} = \frac{p(z_0)}{q'(z_0)}$$

Example. Evaluate for $0 < \alpha < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1 + e^x} dx$$

Proof. We need to take a rectangular contour C_R from $-R$ to R and height $2\pi i$ notice that the above function has a residue at πi . Take

$$f(z) = \frac{\alpha z}{1 + e^z}$$

For every $R > 0$,

$$\int_{C_R} f(z) dz = 2\pi i \text{Res}_{z=\pi i} f(z) = 2\pi i \frac{e^{\alpha \pi i}}{e^{\pi i}} = -2\pi i e^{\alpha \pi i}$$

On the other hand,

$$\int_{C_R} f(z) dz = \int_{-R}^R \frac{e^{\alpha x}}{1+e^x} dx + \int_0^{2\pi} \frac{e^{\alpha(R+iy)}}{1+e^{R+iy}} i dy + \int_R^{-R} \frac{e^{\alpha(x+2\pi i)}}{1+e^{x+2\pi i}} dx + \int_{2\pi}^0 \frac{e^{\alpha(-R+iy)}}{1+e^{-R+iy}} i dy$$

The second term goes to zero as $R \rightarrow \infty$ and

$$-2\pi i e^{\alpha\pi i} = \int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x} dx [1 - e^{2\alpha\pi i}]$$

resulting in

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x} dx = -\frac{2\pi i}{1 - e^{2\alpha\pi i}} = \frac{\pi}{\frac{e^{\alpha\pi i} - e^{-\alpha\pi i}}{2i}} = \frac{\pi}{\sin \alpha\pi}$$

If we further put $t = e^x$, $dx = e^{-x}$, and $dt/t = dx$. We therefore have

$$\int_0^{\infty} \frac{t^{\alpha-1}}{1+t} dt = \frac{\pi}{\sin \alpha\pi}$$

□

LECTURE 14 (NOVEMBER 4TH)

Definition. (Fourier transform of f on $\mathbf{R}$) The Fourier transform of f on $\mathbf{R}$ is defined as

$$\hat{f}(t) = \mathcal{F}(f)(t) = \int_{-\infty}^{\infty} f(x) e^{-itx} dx$$

then

$$f(x) \sim \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{itx} dt$$

for a very huge class of functions f , the above is an equality. If we define

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-2\pi ixt} dx$$

then

$$f(x) \sim \int_{-\infty}^{\infty} \hat{f}(t) e^{2\pi ixt} dt$$

(this still works for f on $\mathbf{R}^n$) We also conventionally use

$$\hat{f}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-itx} dx$$

then

$$f(x) \sim \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt$$

Lastly, we also use

$$\hat{f}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-itx} dx$$

then

$$f(x) \sim \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt$$

The above works for $f \in L^1$, meaning that $\int_{-\infty}^{\infty} |f(x)| dx < \infty$. We can also define the above as

$$\hat{f}(t) = \lim_{A \rightarrow \infty} \int_{-A}^A f(x) e^{-itx} dx$$

when the limits exists.

Example. (Example of transform not being L^1) Consider the L^1 function

$$f(x) = \begin{cases} 1 & -1 < x < 1 \\ 0 & |x| > 1 \end{cases}$$

the Fourier transform is then

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-itx} dx = \int_{-1}^1 e^{-itx} dx = -\frac{1}{it} e^{-itx} \Big|_{-1}^1 = -\frac{1}{it} [e^{-it} - e^{it}] = \frac{2 \sin t}{t}$$

(which is 2 at $t = 0$) Notice that

$$\int_0^{N\pi} \left| \frac{\sin t}{t} \right| dt = \sum_{k=0}^{N-1} \int_{k\pi}^{(k+1)\pi} \left| \frac{\sin t}{t} \right| dt$$

where

$$\int_{k\pi}^{(k+1)\pi} \left| \frac{\sin t}{t} \right| dt \geq \int_{k\pi + \frac{\pi}{6}}^{k\pi + \frac{5\pi}{6}} \left| \frac{\sin t}{t} \right| dt \geq \frac{1}{2(k+1)\pi} \frac{2\pi}{3} = \frac{1}{3(k+1)}$$

and that the result is not L^1 . Is the inverse transform of $g(t) = 2 \sin / t$

$$\lim_{A \rightarrow \infty} \int_{-A}^A g(t) e^{ixt} dt = f(x)$$

is a good question.

Example. Let $a > 0$ and consider the L^1 function

$$f(x) = \frac{1}{x^2 + a^2}$$

and

$$\hat{f}(t) = \int_{-\infty}^{\infty} \frac{e^{-itx}}{x^2 + a^2} dx$$

Proof. Define

$$f(z) = \frac{e^{-itz}}{z^2 + a^2}$$

and use the residue theorem. In the upper hemisphere contour C_R with $R > a$,

$$\int_{C_R} f(z) dz = 2\pi i \operatorname{Res}_{z=ai} f(z) = 2\pi i \frac{e^{-it(ai)}}{2ai} = \frac{\pi}{a} e^{at}$$

Recall that for a simple pole,

$$\operatorname{Res}_{z=z_0} \left(\frac{p(z)}{q(z)} \right) = \frac{p(z_0)}{q'(z_0)}$$

The above is equal to

$$\int_{-R}^R \frac{e^{-itx}}{x^2 + a^2} dx + \int_0^\pi \frac{e^{-itRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} iRe^{i\theta} d\theta$$

The absolute value of the numerator is

$$\left| e^{-itRe^{i\theta}} \right| = \left| e^{-it(R \cos \theta + iR \sin \theta)} \right| = e^{tR \sin \theta}$$

and the second term of the above vanishes at $R \rightarrow \infty$ if the above is bounded and $t \leq 0$.

In the case that $t > 0$, we take the lower hemisphere and the pole is at $-ai$.

$$\int_{C_R} f(z) dz = -2\pi i \operatorname{Res}_{z=-ai} f(z) = 2\pi i \frac{e^{-it(-ai)}}{-2ai} = \frac{\pi}{a} e^{-at}$$

Likewise,

$$\int_{C_R} f(z) dz = \int_{-R}^R \frac{e^{-itx}}{x^2 + a^2} dx + \int_0^{-\pi} \frac{e^{-itRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} iRe^{i\theta} d\theta$$

and

$$\left| e^{-itRe^{i\theta}} \right| = e^{tR \sin \theta} < 1$$

since $-\pi\theta < \theta < 0$. Thus,

$$\hat{f}(t) = \int_{-\infty}^{\infty} \frac{e^{-itx}}{x^2 + a^2} dx = \frac{\pi}{a} e^{-a|t|}$$

which is L^1 . Let $a = 1$ and try to invert the above. If $g(t) = \pi e^{-|t|}$ then

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\infty}^{\infty} g(t) e^{ixt} dt &= \frac{1}{2} \int_{-\infty}^{\infty} e^{-|t|} e^{ixt} dt = \frac{1}{2} \left[\int_{-\infty}^0 e^t e^{ixt} dt + \int_0^{\infty} e^{-t} e^{ixt} dt \right] \\
&= \frac{1}{2} \left[\int_{-\infty}^0 e^{(1+ix)t} dt + \int_0^{\infty} e^{(-1+ix)t} dt \right] \\
&= \frac{1}{2} \left[\frac{e^{(1+ix)t}}{1+ix} \right]_{t=-\infty}^0 + \frac{1}{2} \left[\frac{e^{(-1+ix)t}}{-1+ix} \right]_{t=0}^{\infty} \\
&= \frac{1}{2} \left[\frac{1}{1+ix} \right] + \frac{1}{2} \left[\frac{1}{1-ix} \right] \\
&= \frac{1}{2} \left(\frac{1}{1+ix} + \frac{1}{1-ix} \right) = \frac{1}{1+x^2}
\end{aligned}$$

□

Example. Let $f(x) = e^{-x^2/2}$. Then,

$$\begin{aligned}
\hat{f}(t) &= \int_{-\infty}^{\infty} f(x) e^{-itx} dx = \int_{-\infty}^{\infty} e^{-x^2/2} e^{-itx} dx \\
&= \int_{-\infty}^{\infty} e^{-(x^2+2itx)} dx = \left[\int_{-\infty}^{\infty} e^{-(x+it)^2/2} dx \right] e^{-t^2/2}
\end{aligned}$$

which is $\sqrt{2\pi}$ when $t = 0$. When $t > 0$, we take the rectangular contour with base $2R$ and height $x + it$. By Cauchy,

$$0 = \int_{C_R} e^{-z^2/2} dz = \int_{-R}^R e^{-x^2/2} dx + (\text{II}) + (\text{III}) + (\text{IV})$$

We see that

$$(\text{III}) = \int_R^{-R} e^{-(x+it)^2/2} dt$$

and

$$(\text{II}) = \int_0^t e^{(R+iy)^2/2} i dy$$

We now use the fact that the absolute value of (II) and (III) goes to zero as $R \rightarrow \infty$. We finally have

$$\int_{-\infty}^{\infty} e^{-(x+it)^2/2} dx = \sqrt{2\pi}$$

with

$$\hat{f}(t) = \sqrt{2\pi} e^{-t^2/2}$$

LECTURE 15 (NOVEMBER 6TH)

Recall. We have previously shown the result (for $0 < \alpha < 1$)

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1 + e^x} dx = \frac{\pi}{\sin \alpha \pi}$$

where the change of variables $x = t^\alpha$ gives

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \frac{\pi}{\sin \alpha \pi}$$

Now put $x = t^\beta$ for $0 < \beta < \infty$.

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \int_0^\infty \frac{t^{\alpha\beta-1}}{1+t^\beta} \beta t^{\beta-1} dt = \beta \int_0^\infty \frac{t^{\alpha\beta-1}}{1+t^\beta} dt = \frac{\pi}{\alpha \pi}$$

Take $\alpha = 1/\beta$ such that $\beta > 1$. We get

$$\int_0^\infty \frac{1}{1+t^\beta} dt = \frac{1}{\beta \sin \frac{\pi}{\beta}}$$

for $\beta > 1$.

Recall. We have considered Fourier transform of

$$f(x) = \begin{cases} 1 & -1 < x < 1 \\ 0 & |x| > 1 \end{cases}$$

which was 2 times the sinc function

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-itx} dx = \frac{2 \sin t}{t}$$

We want to check that the inverse transform properly gives the original function.

Example. Find $\lim_{R \rightarrow \infty} \int_{-R}^R \frac{\sin x}{x} e^{itx} dx$. Notice that the kernel is an entire function, and that

$$\int_{-R}^R f(z) dz = \int_{C_R} f(z) dz = \int_{C_R} \frac{e^{iz} - e^{-iz}}{2iz} e^{itz} dz$$

where C_R is defined as contour that avoids the origin. Then,

$$\begin{aligned} \int_{-R}^R f(z) dz &= \int_{C_R} \frac{1}{2i} \left[\frac{e^{i(t+1)z} - e^{i(t-1)z}}{z} \right] dz \\ &= \frac{1}{2i} \left[\int_{C_R} \frac{e^{i(t+1)z}}{z} dz - \int_{C_R} \frac{e^{i(t-1)z}}{z} dz \right] \\ &= \varphi_R(t+1) - \varphi_R(t-1) \end{aligned}$$

where $\varphi_R(t) = \frac{1}{2i} \int_{C_R} \frac{e^{itz}}{z} dz$. Notice that

$$\left| \frac{e^{itz}}{z} \right| = \left| \frac{e^{itRe^{i\theta}}}{Re^{i\theta}} iRe^{i\theta} \right| = |e^{-tR \sin \theta}|$$

To evaluate

$$\varphi_R(t) = \frac{1}{2i} \int_{C_R} \frac{e^{itz}}{z} dz$$

where for $t < 0$, we choose a POSCC so that θ is from 0 to $-\pi$. We find

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{2i} \frac{e^{itz}}{z} dz = 0$$

For $t > 0$, we choose the POSCC $\Gamma_R = C_R \cup \{\text{upper semi circle}\}$. We find

$$\int_{\Gamma_R} \frac{e^{itz}}{z} dz = 2\pi i$$

thus

$$\lim_{R \rightarrow \infty} \int_{C_R} \frac{1}{2i} \frac{e^{itz}}{z} dz = \pi$$

Our goal was

$$\begin{aligned} \lim_{R \rightarrow \infty} \int_{-R}^R \frac{\sin x}{x} e^{itx} dx &= \lim_{R \rightarrow \infty} \int_{C_R} \frac{\sin x}{x} e^{itx} dx = \lim_{R \rightarrow \infty} [\varphi_R(t+1) - \varphi_R(t-1)] \\ &= \lim_{R \rightarrow \infty} \varphi_R(t+1) - \lim_{R \rightarrow \infty} \varphi_R(t-1) \\ &= \begin{cases} \pi & -1 < t < 1 \\ \pi/2 & t = 0 \\ 0 & |t| > 1 \end{cases} \end{aligned}$$

Now, $\hat{f}(t) = 2 \sin t/t$ gives

$$\lim_{R \rightarrow \infty} \int_{-R}^R \hat{f}(t) e^{ixt} dt$$

We can further find that $\lim_{R \rightarrow \infty} \varphi(0) = (i\pi)/2i = \pi/2$.

Theorem. (Big theorem)

- (i) If f and f' are L^1 , $\hat{f}'(t) = it\hat{f}(t)$
- (ii) If $f(x)$ and $xf(x)$ are in L^1 , then

$$(xf(x))^\wedge(t) = i(\hat{f})'(t)$$

and

$$\mathcal{F}(xf(x))(t) = i(\hat{f})'(t)$$

Proof. (I)

$$\hat{f}'(t) = \int_{-\infty}^{\infty} f'(x)e^{-itx} dx = \left[f(x)e^{-itx} \right]_{x=-\infty}^{\infty} + \int_{-\infty}^{\infty} f(x)ite^{-itx} dx$$

However, the first term is zero as

$$\lim_{x \rightarrow \infty} f(x) = f(0) + \int_0^{\infty} f'(x) dx$$

where $\int_0^{\infty} f'(x) dx = \lim_{x \rightarrow \infty} f(x) - f(0)$ and if $\lim_{x \rightarrow \infty} f(x) = \alpha$, $\alpha = 0$. (II)

$$\frac{d}{dt} \hat{f}(t) = \frac{d}{dt} \int_{-\infty}^{\infty} f(x)e^{-ixt} dt = \int_{-\infty}^{\infty} -ixf(x)e^{-ixt} dt = -i \left[xf(x) \right]^{\wedge}(t)$$

□

Example. Last class we saw that for $f(x) = \exp(-x^2/2)$ implies $\hat{f}(t) = \sqrt{2\pi} \exp(-t^2/2)$. Then,

$$f'(x) = -xe^{-x^2/2}$$

so that

$$f'(x) = -xf(x)$$

Applying the Fourier transform,

$$it\hat{f}(t) = -i(\hat{f})'(t)$$

and

$$(\hat{f})'(t) = -t\hat{f}(t)$$

LECTURE 16 (NOVEMBER 11TH)

Theorem. $f \in L^1$ implies that $\int_{-\infty}^{\infty} |f(x)| dx < \infty$. Let $f, f' \in L^1$. Then,

$$(f')^{\wedge}(t) = it\hat{f}(t)$$

and $f, xf(x) \in L^1$ implies that

$$\mathcal{F}(xf(x))(t) = i(\hat{f})'(t)$$

Example. If $f(x) = e^{-ax^2/2}$, $a > 0$, then $f'(x) = -axf(x)$. Take the Fourier transform,

$$it\hat{f}(t) = -ai(\hat{f})'(t)$$

so that $(\hat{f})'(t) = -t\hat{f}(t)/a$ denotes $g(t) = \hat{f}(t)$. Then $g'(t) = -tg(t)/a$ and

$$g(0) = \hat{f}(0) = \int_{-\infty}^{\infty} e^{-ax^2/2} dx = \sqrt{\frac{2\pi}{a}}$$

If I is an open interval containing 0 such that $g(t) > 0$ on I . Then for $t \in I$,

$$\frac{g'(t)}{g(t)} = -\frac{t}{a}$$

implies that

$$\ln g(t) = -\frac{t^2}{2a} + C$$

and $g(t) = C_1 e^{-t^2/2a}$. Put $t = 0$, and

$$g(t) = \sqrt{\frac{2\pi}{a}} e^{-t^2/2a}$$

on I . If $I \neq \mathbf{R}$, then there is an endpoint t_0 of I such that $g(t_0) \leq 0$. Put

$$g(t_0) = \lim_{t \rightarrow t_0} g(t) = \lim_{t \rightarrow t_0} \sqrt{\frac{2\pi}{a}} e^{-t^2/2a} = \sqrt{\frac{2\pi}{a}} e^{-t_0^2/2a} > 0$$

which is a contradiction.

$$\mathcal{F}\left(e^{-ax^2/2}\right)(t) = \sqrt{\frac{2\pi}{a}} e^{-t^2/2a}$$

for $a > 0$.

Definition. (Convolution of f and g) Let

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y)g(y) dy$$

wherever it is defined.

Theorem. If $f, g \in L^1$ and $h = f * g$, then $h \in L^1$.

Example. Let

$$f(x) = \begin{cases} 1/\sqrt{x} & (0, 1) \\ 0 & \text{elsewhere} \end{cases} \quad g(x) \in L^1$$

also,

$$f(x)g(x) = \begin{cases} 1/x & (0, 1) \\ 0 & \text{elsewhere} \end{cases}$$

which is not L^1 . Thus, L^1 isn't closed under multiplication.

Lemma.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dy \, dx$$

if

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x, y)| \, dx \, dy < \infty$$

Also, if $f(x, y) \geq 0$ for all x, y , then the above also holds.

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} |h(x)| \, dx &= \int_{-\infty}^{\infty} (f * g(x)) \, dx = \int_{-\infty}^{\infty} \left| \int_{-\infty}^{\infty} f(x-y)g(y) \, dy \right| \, dx \\ &\leq \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x-y)||g(y)| \, dy \, dx \leq \int_{-\infty}^{\infty} |g(y)| \, dy \int_{-\infty}^{\infty} |f(x-y)| \, dx < \infty \end{aligned}$$

□

Theorem. If $f, g \in L^1$, then $f * g = g * f$.

$$(f * g)(t) = \int_{-\infty}^{\infty} f(x-y)g(y) \, dy = \int_{-\infty}^{\infty} f(t)g(x-t)(-dt) = \int_{-\infty}^{\infty} g(x-t)f(t) \, dt = (g * f)(x)$$

by $x - y = t$ and $dy = -dt$. Thus,

$$(f * g)(x) = \int_{-\infty}^{\infty} g(x-y)f(y) \, dy$$

Remark. The meaning of $f * g$ is the weighted average of f using g . For $a > 0$, consider

$$g_a(x) = \begin{cases} 1/2a & -a < x < a \\ 0 & \text{elsewhere} \end{cases}$$

then,

$$g_a(x) \geq 0 \quad \& \quad \int_{-\infty}^{\infty} g_a(x) \, dx = 1 \quad \& \quad g_a(-x) = g_a(x)$$

then

$$(f * g_a)(x) = \int_{-\infty}^{\infty} g_a(x-y)f(y) \, dy$$

where

$$g_a(x-y) = \begin{cases} 1/2a & x-a < y < x+a \\ 0 & \text{elsewhere} \end{cases}$$

Notice how

$$(f * g)(x) = \frac{1}{2a} \int_{x-a}^{x+a} f(y) \, dy$$

taken to $a \rightarrow 0^+$ leaves us with $(f * g_a)(x) \rightarrow f(x)$ if $f(x)$ is continuous at x .

$$\lim_{a \rightarrow 0^+} (f * g_a)(x) = \frac{1}{2}(f(x_+) + f(x_-))$$

Remark. If $\phi \geq 0$,

$$\int_{-\infty}^{\infty} \phi(x) dx = 1$$

and $\phi(-x) = \phi(x)$, then

$$\phi_\varepsilon(x) = \frac{1}{\varepsilon} \phi\left(\frac{x}{\varepsilon}\right)$$

with $\varepsilon > 0$ satisfies

$$\int_{-\infty}^{\infty} \phi_\varepsilon(x) dx = \int_{-\infty}^{\infty} \frac{1}{\varepsilon} \phi\left(\frac{x}{\varepsilon}\right) dx = \int_{-\infty}^{\infty} \phi(t) dt = 1$$

and

$$\lim_{\varepsilon \rightarrow 0^+} (f * \phi_\varepsilon)(x) = \frac{1}{2}(f(x_+) + f(x_-))$$

For example, we have the Cauchy distribution

$$\phi(x) = \frac{1}{\pi} \frac{1}{1 + x^2}$$

and the normal distribution

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

which is the smoothing function. Notice that

$$(f * \phi)(x) = (\phi * f)(x) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x-y)^2/2} f(y) dy$$

(the exponential function is $C^\infty(\mathbf{R})$ and is nice)

$$(f * \phi_\varepsilon)(x) = \int_{-\infty}^{\infty} \frac{1}{\varepsilon \sqrt{2\pi}} e^{-|x-y|^2/2\varepsilon^2} f(y) dy$$

which implies

$$\frac{1}{2}|f(x_+) + f(x_-)|$$

and $\varepsilon \rightarrow 0$. The above is exactly $f(x)$ on every norm. Note that

$$\mathcal{F}\left(e^{-ax^2/2}\right)(t) = \sqrt{\frac{2\pi}{a}} e^{-t^2/2a}$$

and $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$. Then for $f \in L^1$,

$$(f * \phi_\varepsilon)(x) = \int_{-\infty}^{\infty} \frac{1}{\varepsilon \sqrt{2\pi}} e^{-|x-y|^2/2\varepsilon^2} f(y) dy$$

with

$$\phi_\varepsilon(x) = \frac{1}{\varepsilon} \phi\left(\frac{x}{\varepsilon}\right) = \frac{1}{\varepsilon\sqrt{2\pi}} e^{-x^2/2\varepsilon^2}$$

Theorem. We want to verify the equality

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt = \frac{1}{2} (f(x_+) + f(x_-))$$

if $f \in L^1$ and is piecewise continuous.

Proof. Indeed,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) e^{-ity} dy e^{ixt} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) e^{i(x-y)t} dy dt$$

This is a dead lock ($|e^{i(x-y)t}| = 1$ and is not L^1 with respect to t). Instead,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{-\varepsilon^2 t^2/2} e^{ixt} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(y) e^{-ity} dy e^{-\varepsilon^2 t^2/2} e^{ixt} dt$$

and

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) e^{-\varepsilon^2 t^2/2} e^{i(x-y)t} dy dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(y) \int_{-\infty}^{\infty} e^{-\varepsilon^2 t^2/2} e^{-i(y-x)t} dt dy$$

where the most right integral satisfies

$$\mathcal{F}\left(e^{-\varepsilon^2 t^2/2}\right)(y-x) = \frac{\sqrt{2\pi}}{\varepsilon} e^{-(y-x)^2/2\varepsilon^2}$$

resulting in

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} f(y) \frac{\sqrt{2\pi}}{\varepsilon} e^{-(y-x)^2/2\varepsilon^2} dy = (f * \phi_\varepsilon)(x) \rightarrow \frac{1}{2} (f(x_+) + f(x_-))$$

as $\varepsilon \rightarrow 0$ and

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{-\varepsilon^2 t^2/2} e^{itx} dt \rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{itx} dt$$

as $\varepsilon \rightarrow 0$. □

LECTURE 17 (NOVEMBER 13TH)

Recall. Consider $f \in L^1$ be piecewise continuous, and define

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

We define

$$\phi_\varepsilon(x) = \frac{1}{\varepsilon} \phi_\varepsilon\left(\frac{x}{\varepsilon}\right) = \frac{1}{\varepsilon\sqrt{2\pi}} e^{-\frac{x^2}{2\varepsilon^2}}$$

this implies that

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) e^{itx} dt = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\hat{f}(t) e^{-\varepsilon^2 t^2 / 2} \right) e^{ixt} dx = \lim_{\varepsilon \rightarrow 0} (f * \phi_\varepsilon)(x) = \frac{1}{2} (f(x^+) + f(x^-))$$

Note that $\phi(x)$ is in the Schwartz class, satisfying

$$\sup_{x \in \mathbf{R}} \left(x^k \phi^{(n)}(x) \right) < \infty$$

for all $k, n \geq 0$.

Definition. (Schwartz class $\mathcal{S}$) A function is in the Schwartz class $f \in \mathcal{S}$ if and only if $f \in C^\infty(\mathbf{R})$ and

$$\sup_{x \in \mathbf{R}} |x^k f^{(n)}(x)| < \infty$$

If $f \in \mathcal{S}$ then $(\hat{f})^\vee = f$ where

$$g^\vee(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{ixt} dt$$

Thus, $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ defined by $\mathcal{F}(f) = \hat{f}$ is an isomorphism.

Recall. Recall that

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y)g(y) dy = \int_{-\infty}^{\infty} f(y)g(x-y) dy = (g * f)(x)$$

if $f, g \in L^1$ then $f * g \in L^1$.

Theorem. If $f, g \in L^1$, then $(f * g)^\wedge(t) = \hat{f}(t)\hat{g}(t)$.

Proof.

$$\begin{aligned} (f * g)^\wedge(t) &= \int_{-\infty}^{\infty} (f * g)(x) e^{-ixt} dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-y)g(y) dy e^{-ixt} dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-y)g(y) e^{-ixt} dy dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-y) e^{-i(x-y)t} g(y) e^{-iyt} dx dy \\ &= \int_{-\infty}^{\infty} g(y) e^{-iyt} dy \int_{-\infty}^{\infty} f(x-y) e^{-i(x-y)t} dx \\ &= \hat{g}(t) \hat{f}(t) \end{aligned}$$

□

Theorem. (Heat equation) Let's assume $u(x, t) \in \mathcal{S}$ for every fixed t .

$$\begin{cases} u_{xx} = u_t & t > 0 \\ u(x, 0) = f(x) \end{cases}$$

Define

$$g(w, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ixw} dx$$

which is the Fourier transform of $u(x, t)$. Recall that $\hat{f}'(t) = it\hat{f}(t)$.

$$\int_{-\infty}^{\infty} u_t(x, t) e^{-ixw} dx = \frac{\partial g}{\partial t}(w, t)$$

Together this implies that in the Fourier space,

$$-w^2 g = \frac{\partial g}{\partial t}$$

and the initial condition becomes

$$g(w, 0) = \hat{f}(w)$$

Once fixing w , we see that the above is an ordinary differential equation with variable t .

$$g(w, t) = \hat{f}(w) e^{-iw^2 t}$$

Last class, we've also learned

$$\mathcal{F}(e^{-ax^2/2})(w) = \sqrt{\frac{2\pi}{a}} e^{-w^2/2a}$$

Now, for fixed $t > 0$, put $a = 1/2t$,

$$\mathcal{F}\left(\sqrt{\frac{1}{4\pi t}} e^{-x^2/4t}\right) = e^{-w^2 t}$$

Now define $h_t(x) = \sqrt{1/4\pi t} \exp(-x^2/4t)$, then $\hat{h}_t(w) = \exp(-w^2 t)$.

$$g(w, t) = \hat{u}(w, t) = \hat{f}(w) \hat{h}_t(w)$$

By the inversion theorem, together with

$$(f * h_t)^\wedge = \hat{f} \hat{h}_t$$

we have $u(x, t) = (f * h_t)(x)$

$$u(x, t) = \int_{-\infty}^{\infty} f(y) \frac{1}{\sqrt{2\pi t}} e^{-(x-y)^2/4t} dy$$

Recall. Let

$$g(t) = \begin{cases} e^{-1/t} & t > 0 \\ 0 & t \leq 0 \end{cases}$$

then $g \in C^\infty(\mathbf{R})$. Also, let

$$h(t) = \begin{cases} e^{-1/(1-t^2)} & -1 < t < 1 \\ 0 & |t| \geq 1 \end{cases}$$

then $h \in C_c^\infty(\mathbf{R})$, meaning that C^∞ with a compact support. $f \in C_c(\mathbf{R})$ with support K when $f(x) = 0$ if $x \in K^c$. Finally, define $\psi : \mathbf{R} \rightarrow \mathbf{R}$ by

$$\psi(x) = \frac{h(x)}{\int_{-1}^1 h(t) dt}$$

then $\psi \in C_c^\infty(\mathbf{R})$ with support $[-1, 1]$ and $\int_{-\infty}^{\infty} \psi(x) dx = 1$. $\psi \in C_c^\infty(\mathbf{R})$ satisfies $\psi \geq 0$, $\psi(-x) = \psi(x)$, $\int_{-\infty}^{\infty} \psi(x) dx = 1$. Define

$$\psi_\varepsilon(x) = \frac{1}{\varepsilon} \psi\left(\frac{x}{\varepsilon}\right)$$

and $f \in L^1(\mathbf{R})$ and $f * \psi_\varepsilon \in C^\infty(\mathbf{R})$ with $f * \psi_\varepsilon \rightarrow f$.

Theorem. If f and g have a compact support, then $f * g$ also has a compact support.

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y)g(y) dy$$

has a compact support.

Corollary. If $f \in L^1$ with a compact support, then $f * \psi_\varepsilon \in C_c^\infty(\mathbf{R}) \subset \mathcal{S}$.

Recall. If f satisfies $\int_{\mathbf{R}} |f|^p dx < \infty$ for $1 \leq p < \infty$, then for every $\varepsilon > 0$ there is $h \in C_c(\mathbf{R})$ such that $\int_{\mathbf{R}} |f - h|^p dx < \varepsilon$. This tells us that $C_c(\mathbf{R})$ is dense in $L^p(\mathbf{R})$ for $1 \leq p < \infty$.

Example. Let f satisfy $\int_{-\infty}^{\infty} |f(y)|^2 dy < \infty$. Define $g(x) = \overline{f(-x)}$. For $f \in \mathcal{S}$,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) dt = \left(\hat{f}\right)^\vee(0) = f(0)$$

and $\int_{-\infty}^{\infty} \hat{f}(t) dt = \sqrt{2\pi} f(0)$. Then,

$$\begin{aligned} (f * g)(0) &= \int_{-\infty}^{\infty} g(0-y) f(y) dy \\ &= \int_{-\infty}^{\infty} \overline{f(y)} f(y) dy = \int_{-\infty}^{\infty} |f(y)|^2 dy \end{aligned}$$

Thus,

$$\begin{aligned} \int_{-\infty}^{\infty} |f(y)|^2 dy &= (f * g)(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (f * g)^{\wedge}(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(t) \hat{g}(t) dt \\ \hat{g}(t) &= \int_{-\infty}^{\infty} g(x) e^{-ixt} dt = \int_{-\infty}^{\infty} \overline{f(-x)} e^{-ixt} dx = \int_{-\infty}^{\infty} \overline{f(y)} e^{iyt} (-dy) \\ &= \overline{\int_{-\infty}^{\infty} f(y) e^{-iyt} dy} = \overline{\hat{f}(t)} \end{aligned}$$

by inversion theorem if $f, g \in \mathcal{S}$. In conclusion,

$$\int_{-\infty}^{\infty} |f(y)|^2 dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(t)|^2 dt$$

holds for every f with $\int_{-\infty}^{\infty} |f(y)|^2 dy < \infty$. Plancherel theorem.

LECTURE 18

Recall. We have previously learned the Schwartz class $\mathcal{S}$ and that it is $\mathcal{S}$ is dense in L^1 . This implied that for $f \in L^1$, there is $\{f_n\}$ in $\mathcal{S}$ such that

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} |f_n - f| dx = 0$$

Also, if $f \in \mathcal{S}$, $\hat{f} \in \mathcal{S}$.

Theorem. (Riemann Lebesgue lemma) If $f \in L^1$, then

$$\lim_{|t| \rightarrow \infty} \hat{f}(t) = 0$$

Proof. Take any $\varepsilon > 0$, we have to find $M > 0$ satisfying if $|t| > M$, then $|\hat{f}(t)| < \varepsilon$. We use the fact that $\mathcal{S}$ is dense in L^1 and that $\hat{f}(t) = \hat{f}_n(t) + (\hat{f}(t) - \hat{f}_n(t))$ where $f_n \in \mathcal{S}$. For $f \in L^1$, there is $f_N \in \mathcal{S}$ such that

$$\int_{-\infty}^{\infty} |f - f_N| dx < \frac{\varepsilon}{2}$$

Since $\hat{f}_N \in \mathcal{S}$, there is $M > 0$ such that if $|t| > M$ then $|\hat{f}_N(t)| < \varepsilon/2$. And

$$\hat{f}_N(t) - \hat{f}(t) = \int_{-\infty}^{\infty} (f_N(x) - f(x)) e^{-itx} dx$$

so that

$$|\hat{f}_N(t) - \hat{f}(t)| \leq \int_{-\infty}^{\infty} |f_N(x) - f(x)| dx < \frac{\varepsilon}{2}$$

Hence, if $|t| > M$ then

$$|\hat{f}(t)| \leq |\hat{f}_N(t)| + |\hat{f}(t) - \hat{f}_N(t)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

□

Corollary. If $f \in L^1$, then $\hat{f}$ is continuous on $\mathbf{R}$. Take $\{f_n\}$ in $\mathcal{S}$ such that

$$\int_{-\infty}^{\infty} |f_n - f| dx \rightarrow 0$$

then for every $t \in \mathbf{R}$

$$|\hat{f}_n(t) - \hat{f}(t)| \leq \int_{-\infty}^{\infty} |f_n(x) - f(x)| dx$$

That is, $\hat{f}_n \rightarrow \hat{f}$ uniformly on $\mathbf{R}$.

Definition. (Laplace transform of f) Let $f(t) = 0$ if $t < 0$, and continuous for $f \geq 0$. Lastly, let there exist $M, k, t_0 > 0$ such that

$$|f(t)| \leq M e^{kt}$$

for every $t > t_0$. For such f , the Laplace transform F of f is defined as

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

where $\operatorname{Re}(s) > k$ and $s \in \mathbf{C}$, analytic on the half plane $\operatorname{Re}(s) > k$. Indeed, if $s = \sigma + iw$ (where $\sigma > k$) then

$$F(s) = \int_0^{\infty} f(t) e^{-\sigma t} e^{-iwt} dt$$

which is $\hat{g}(w)$ where $g(t) = f(t) e^{-\sigma t}$ and is L^1 . We now apply the inversion theorem on

$$\hat{g}(w) = \int_0^{\infty} f(t) e^{-\sigma t} e^{-iwt} dt = \int_0^{\infty} f(x) e^{-\sigma x} e^{-iwx} dx$$

Then,

$$f(t) e^{-\sigma t} = g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{g}(w) e^{itw} dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) e^{-\sigma x} e^{-iwx} dx e^{itw} dw$$

Hence,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_0^{\infty} f(x) e^{-(\sigma+iw)x} dx e^{(\sigma+iw)t} dw$$

Putting pack $s = \sigma + iw$ ($dw = ds/i$),

$$f(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \int_0^{\infty} f(x) e^{-sx} dx e^{st} ds = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} F(s) e^{st} ds$$

for $t > 0$.

Example. If $f(t) = 1 - e^{-t}$ for $t \geq 0$, zero elsewhere. Then

$$F(s) = \int_0^{\infty} (1 - e^{-t}) e^{-st} dt = \int_0^{\infty} e^{-st} dt - \int_0^{\infty} e^{-(s+1)t} dt = \frac{1}{s} - \frac{1}{s+1} = \frac{1}{s(s+1)}$$

Let's try the inversion theorem!

$$f(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s(s+1)} ds$$

and we the left hemisphere contour $C_R \cup \{x = \sigma\}$. For the right edge, we use $s = \sigma + iy$ ($-R \leq y \leq R$). For the round circumference, $s = \sigma + Re^{i\theta}$ with $\pi/2 \leq \theta \leq 3\pi/2$. Let $f(s) = e^{st}/s(1+s)$ Always,

$$\lim_{R \rightarrow \infty} \int_{C_R} f(s) ds = 0$$

must hold. In this particular case,

$$\int_{C_R} f(s) ds \rightarrow 0$$

as $R \rightarrow \infty$. Thus,

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s(1+s)} ds = 2\pi i \frac{1}{2\pi i} (1 - e^{-t}) = 1 - e^{-t}$$

Recall. Notice that

$$\int_0^{\pi} e^{-R \sin \theta} d\theta = 2 \int_0^{\pi/2} e^{-R \sin \theta} d\theta$$

and that $2\theta/\pi \leq \sin \theta \leq \theta$.

$$\int_0^{\pi/2} e^{-R \sin \theta} d\theta \leq \int_0^{\pi/2} e^{-R\theta/2\pi} d\theta = \frac{\pi}{2R} (1 - e^{-R}) \leq \frac{\pi}{2R}$$

and

$$\int_0^{\pi} e^{-R \sin \theta} d\theta \leq \frac{\pi}{R}$$

Example. Find

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s^2+1} ds = \operatorname{Res}_{z=i} f + \operatorname{Res}_{z=-i} f =$$

where $f(s) = e^{st}/(s^2+1)$. Notice that

$$\operatorname{Res}_{z=\pm i} f = \frac{p(\pm i)}{q'(\pm i)}$$

where $p(s) = e^{st}$ and $q(s) = s^2+1$. We thus have

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{st}}{s^2+1} ds = \frac{e^{it}}{2i} - \frac{e^{-it}}{2i} = \sin t$$

Example. Find

$$\frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{2s^3 e^{st}}{s^4-4} ds$$

Note that $s^4-4=0$ implies $s = \pm\sqrt{2}, \pm i\sqrt{2}$.

$$\operatorname{Res}_{z=\alpha} f = \frac{2\alpha^3 e^{\alpha t}}{4\alpha^3} = \frac{e^{\alpha t}}{2}$$

and

$$\frac{e^{t\sqrt{2}}}{2} + \frac{e^{-t\sqrt{2}}}{2} + \frac{e^{-it\sqrt{2}}}{2} + \frac{e^{it\sqrt{2}}}{2} = \cosh \sqrt{2}t + \cos \sqrt{2}t$$

LECTURE 19 (NOVEMBER 20TH)

Remark. Consider the function

$$f(z) = \pi \cot \pi z = \frac{\pi \cos \pi z}{\sin \pi z}$$

which has a simple pole at each $n \in \mathbf{Z}$.

$$\operatorname{Res}_{z=n} f(z) = \frac{\pi \cos n\pi}{(\sin \pi z)'_{z=n}} = \frac{\pi \cos n\pi}{\pi \cos n\pi} = 1$$

Consider the square contour C_N and we find

$$\pi \cot \pi z = \frac{1}{z} \left(1 - \frac{\pi^2 z}{3} - \frac{\pi^4 z^4}{45} - \dots \right)$$

Example. We can use the above function to evaluate

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \xi(4)$$

Observe how

$$\int_{C_N} \frac{\pi \cot \pi z}{z^4} dz \rightarrow 0$$

as $N \rightarrow \infty$ due to parametrisation. This is in turn, due to the residue theorem,

$$2\pi i \sum_{k=-N}^N \operatorname{Res}_{z=k} \left[\frac{\pi \cot \pi z}{z^4} \right] = 2\pi i \left[2 \sum_{k=1}^N \frac{1}{k^4} + \operatorname{Res}_{z=0} \frac{\pi \cot \pi z}{z^4} \right]$$

and we expect the residue in the bracket to be

$$\operatorname{Res}_{z=0} \frac{\pi \cot \pi z}{z^4} = -\frac{\pi^4}{45}$$

and

$$\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$$

Example. For $a \notin \mathbf{Z}$, evaluate

$$\sum_{n=-\infty}^{\infty} \frac{1}{(z+a)^2}$$

Proof.

$$\int_{C_N} \frac{\pi \cot \pi z}{(n+a)^2} dz = 2\pi i \left[\sum_{k=-N}^N \operatorname{Res}_{z=k} \left(\frac{\pi \cot \pi z}{(z+a)^2} \right) + \operatorname{Res}_{z=-a} \left(\cdot \right) \right]$$

If f has a pole of order 2 at z_0 , then

$$\operatorname{Res}_{z=z_0} f(z) = \frac{1}{(k-1)!} \lim_{z \rightarrow z_0} \left[(z-z_0)^k f(z) \right]^{(k-1)}$$

Thus

$$\operatorname{Res}_{z=-a} \frac{\pi \cot \pi z}{(z+a)^2} = \lim_{z \rightarrow -a} [\pi \cot \pi z]' = [\pi(-\csc^2 \pi z)\pi]_{z=-a} = -\frac{\pi^2}{\sin^2 a\pi}$$

Thus,

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+a)^2} = \frac{\pi^2}{\sin^2 a\pi}$$

□

Example. Define

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + a^2}$$

for $a > 0$. Use

$$\int_{C_N} \frac{\pi \cot \pi z}{n^2 + a^2} dz \rightarrow 0$$

as $N \rightarrow \infty$. The above is equal to

$$2\pi i \left[\sum_{k=-N}^N \left(\cdot \right) + \operatorname{Res}_{z=ai} \left(\cdot \right) + \operatorname{Res}_{z=-ai} \left(\cdot \right) \right]$$

We have

$$\frac{1}{2} \left(\frac{\pi \cosh \pi a}{a \sinh \pi a} - \frac{1}{a^2} \right)$$

Theorem. For $z \notin \mathbf{Z}$,

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}$$

For $w \in \mathbf{Z}$,

$$\int_{C_N} \frac{\pi \cot \pi z}{z(z-w)} dz = 2\pi i \left[\sum_{0 \neq k=-N}^N \operatorname{Res}_{z=k} \frac{\pi \cot \pi z}{z(z-w)} + \operatorname{Res}_{z=0} \frac{\pi \cot \pi z}{z(z-w)} + \operatorname{Res}_{z=w} \left(\cdot \right) \right]$$

which goes to 0 as $N \rightarrow \infty$. The first residue is equal to

$$\frac{1}{k(k-w)}$$

and the last is equal to

$$\frac{\pi \cot \pi w}{w}$$

For the second, we use partial fractions,

$$\frac{1}{z(z-w)} = -\frac{1}{w} \left(\frac{1}{z} - \frac{1}{z-w} \right)$$

and we have $-1/w^2$ for the second one. Gathering the results,

$$\int_{C_N} \frac{\pi \cot \pi z}{z(z-w)} dz = 2\pi i \left[\sum_{0 \neq k=-N}^N \frac{1}{k(k-w)} - \frac{1}{w^2} + \frac{\pi \cot \pi w}{w} \right]$$

Notice, however,

$$\sum_{k=1}^N \frac{1}{k(k-w)} + \sum_{k=-N}^{-1} \frac{1}{k(k-w)} = \sum_{k=1}^N \frac{1}{k} \left(\frac{1}{k-w} + \frac{1}{k+w} \right) = \sum_{k=1}^N \frac{1}{k} \frac{2k}{k^2 - w^2} = \sum_{k=1}^N \frac{2}{k^2 - w^2}$$

and

$$\pi \cot \pi w = \frac{1}{w} + \sum_{k=1}^{\infty} \frac{2w}{w^2 - k^2}$$

Example. (Reverse parametrisation)

$$\int_0^{2\pi} \frac{1}{1 + a \cos \theta} d\theta$$

with $0 < a < 1$. We do this through paramterising in the opposite direction, $C : |z| = 1$ and $z = e^{i\theta}$ ($0 \leq \theta \leq 2\pi$).

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + z^{-1}}{2}$$

and $dz = ie^{i\theta} d\theta = iz d\theta$ implying $d\theta = dz/iz$. So,

$$\int_0^{2\pi} \frac{d\theta}{1 + a \cos \theta} = \int_{|z|=1} \frac{1}{1 + a \left(\frac{z + z^{-1}}{2} \right)} \frac{dz}{iz} = \frac{1}{ai} \int_{|z|=1} \frac{2 dz}{z^2 + \frac{2z}{a} + 1}$$

now, $z^2 + 2z/a + 1 = (z - z_1)(z - z_2)$ and

$$z_1 = -\frac{1}{a} + \sqrt{\frac{1}{a^2} - 1} = \frac{-1 + \sqrt{1 - a^2}}{a} \quad \& \quad z_2 = \frac{-1 - \sqrt{1 - a^2}}{a}$$

where the second is outside $|z| = 1$.

$$z_1 - z_2 = \frac{2\sqrt{1 - a^2}}{a}$$

and

$$2\pi i \frac{1}{ai} \operatorname{Res}_{z=z_1} \left(\cdot \right) = 2\pi i \frac{1}{ai} \frac{2}{z_1 - z_2} = \frac{2\pi}{\sqrt{1 - a^2}}$$

Example. Let's see another example of reverse parametrisation. For $n \in \mathbf{N}$, evaluate

$$\int_0^{2\pi} \sin^{2n} t dt = \int_{|z|=1} \left(\frac{z - z^{-1}}{2i} \right)^{2n} \frac{dz}{iz} = \frac{1}{i(2i)^{2n}} \int_{|z|=1} (z - z^{-1})^{2n} z^{-1} dz$$

Let $z = e^{it}$ for $0 \leq t \leq 2\pi$.

$$\sin t = \frac{z - z^{-1}}{2i}$$

with $dt = dz/iz$. We only need to find the coefficient of z^{-1} .

$$(z - z^{-1})^{2n} = \sum_{k=0}^{2n} {}^{2n}C_k z^k (-z^{-1})^{2n-k}$$

Finding the constant coefficient,

$$2\pi i \frac{1}{i} \frac{1}{(2i)^{2n}} {}^{2n}C_n (-1)^n = 2\pi \frac{(2n)!}{2^{2n} (n!)^2}$$

LECTURE 20 (NOVEMBER 25TH)

Remark. Let f be an entire function. The zero set of f is denoted as Z_f . If A is an infinite subset of $\mathbf{C}$ and $Z_f = A$ for some entire function f , then A must not have a limit point. If $A = \{a_n\} \subset \mathbf{C}$ with $\lim_{n \rightarrow \infty} |a_n| = \infty$, then is there an entire function f satisfying $A = Z_f$? The answer is yes.

Example. Let $A = \{a_1, \dots, a_n\}$ be a finite zero set. Then $f(z) = (z - a_1) \dots (z - a_n)$ and $A = Z_f$.

Proposition. If there are two entire functions f and g such that $Z_f = Z_g$, then $\frac{f}{g}$ is entire with no zeros. Since $\mathbf{C}$ is simply connected, there exists an entire function h such that $\frac{f}{g} = e^h$. That is, $f = ge^h$.

Example. An entire function f such that $Z_f = \mathbf{Z}$.

$$f(z) = \sin \pi z$$

How do we find an entire function f such that $Z_f = \{2, 3, 4, 5, 6, \dots\}$? Let's use the notation

$$\prod_{n=1}^N a_n = a_1 a_2 \dots a_N$$

and write

$$f(z) = \left(1 - \frac{z}{2}\right) \left(1 - \frac{z}{3}\right) \left(1 - \frac{z}{4}\right)$$

This function is zero at 1, and our conjecture is false. The idea is to find a multiplier $e^{g(z)}$ to take this property away!

$$f(z) = \prod_{n=2}^{\infty} \left(1 - \frac{z}{n}\right) e^{g(z)}$$

Lemma. If $a_n \geq 0$, then

$$\prod_{n=1}^{\infty} (1 + a_n)$$

converges if and only if

$$\sum_{n=1}^{\infty} a_n$$

converges.

Proof. If $x \geq 0$,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \geq 1 + x$$

□

Proof.

$$\sum_{n=1}^N a_n \leq \prod_{n=1}^N (1 + a_n) \leq \prod_{n=1}^N e^{a_n} = e^{\sum_{n=1}^N a_n}$$

using the comparison test. $\square$

Theorem. (Big theorem I) Let $a_n \in \mathbf{C}$. If $\sum_{n=1}^{\infty} |a_n| < \infty$, then $\prod_{n=1}^{\infty} (1 + a_n)$ converges and if $\prod_{n=1}^{\infty} (1 + a_n) = 0$ then $a_n = -1$ for some $n \in \mathbf{N}$.

Proof. Since $\sum_{n=1}^{\infty} |a_n|$ converges, there is $K \in \mathbf{N}$ such that if $n \geq K$, then $|a_n| < \frac{1}{2}$. Then

$$\prod_{n=k}^N (1 + a_n) = \prod_{n=k}^N e^{\log(1+a_n)} = e^{\sum_{n=k}^N \log(1+a_n)}$$

Notice that

$$\sum_{n=k}^N |\log(1 + a_n)| \leq \sum_{n=k}^N 2|a_n|$$

by lemma. Thus,

$$\sum_{n=k}^{\infty} \log(1 + a_n)$$

converges to some $A \in \mathbf{C}$.

$$\prod_{n=k}^N (1 + a_n) = e^A \neq 0$$

$\square$

Lemma. We denote $\log z$ for $z \neq 0$ and $-\pi < \arg z < \pi$.

$$\log z = \ln |z| + i \arg z$$

If $|z| < \frac{1}{2}$, then $\log(1 + z)$ satisfies $|\log(1 + z)| \leq 2|z|$.

Proof. Since $\log(1 + z)$ is analytic in $|z| < 1$,

$$\log(1 + z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4}$$

for $|z| < 1$. Thus, if $|z| < \frac{1}{2}$,

$$|\log(1 + z)| = \left| z - \frac{z^2}{2} + \frac{z^3}{3} + \dots \right| \leq |z| \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right) = 2|z|$$

$\square$

Theorem. (Big theorem II) Let $\{f_n\}$ be analytic in a domain D . If there is $\{M_n\}$ with $M_n \geq 0$ satisfying $\sum_{n=1}^{\infty} M_n < \infty$ and $|f_n(z) - 1| \leq M_n$ for all $z \in D$, then

$$f(z) = \prod_{n=1}^{\infty} f_n(z)$$

is analytic in D since $\prod_{n=1}^N f_n(z)$ converges uniformly on every compact subset.

Proof. There exists $n \geq k$ such that $|f_n(z) - 1| < \frac{1}{2}$ for all $z \in D$.

$$\prod_{n=k}^N f_n(z) = \prod_{n=k}^N e^{\log(f_n(z))} = e^{\sum \log(f_n(z))} \leq e^{\sum M_n}$$

as $|\log(f_n(z) - 1)| \leq M_n$. □

Lemma. If f_n converges uniformly and g is uniformly continuous,

$$g \circ f_n$$

converges uniformly.

Example. Prove that $\prod_{n=1}^{\infty} \left(1 - \frac{z}{n^2}\right)$ is an entire function with $Z_f = \{n^2 \mid n \in \mathbf{N}\}$.

Proof. Let $D = D(0, R)$. Take $f_n(z) = z/n^2$ and $M_n = R/n^2$ we can conclude that $f(z)$ is analytic in D . □

LECTURE 21 (NOVEMBER 27TH)

Recall. Let f_n be analytic in a domain D . Let

$$\sum_{n=1}^{\infty} |f_n(z) - 1| \leq M_n$$

for all $z \in D$ where $\sum_{n=1}^{\infty} M_n < \infty$. Then,

$$f(z) = \prod_{n=1}^{\infty} f_n(z)$$

is analytic in D , and $f(z_0) = 0$ if and only if $f_n(z_0) = 0$ for some n .

Proof. There is a K such that $M_n < \frac{1}{2}$ for all $n \geq K$. Thus, if $n \geq K$, then $|f_n(z) - 1| < \frac{1}{2}$ for all $z \in D$ ($\frac{1}{2} \leq |f_n(z)| < \frac{3}{2}$ for all $z \in D$ if $n \geq K$). Thus, if $n \geq K$, then $\log f_n$ is analytic in D .

$$\prod_{n=k}^N f_n(z) = \prod_{n=k}^N e^{\log f_n(z)} = e^{\sum_{n=k}^N \log f_n(z)} \rightarrow e^{F(z)}$$

as $|\log f_n(z)| = |1 + \log f_n(z) - 1| \leq 2|f_n(z) - 1| \leq 2M_n$. If K is a compact subset of D , then $F(K)$ is compact and $g(z) = e^z$ is uniformly continuous on $F(K)$, which tells us that e^F is analytic in D . $\square$

Example. The function

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n^2}\right)$$

is an entire function with $Z_f = \{n^2 \mid n \in \mathbf{N}\}$. Let $D = D(0, R)$ and $f_n(z) = 1 - z/n^2$. Then,

$$\sum_{n=1}^{\infty} |1 - f_n(z)| \leq \sum_{n=1}^{\infty} \frac{R}{n^2}$$

setting $R/n^2 = M_n$, f is analytic on $D(0, R)$. Since R is arbitrary, f is entire.

Example. Let $0 \leq a_n < 1$. Then if $\sum_{n=1}^{\infty} a_n < \infty$, then $\prod_{n=1}^{\infty} (1 - a_n) > 0$. Put $f_n(z) = 1 - a_n$ constant.

Proposition. Let

$$A = \{1, 2, 3, 4, \dots\}$$

$$B = \{1, 4, 9, 25, \dots\}$$

$$C = \{1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \dots\}$$

$$D = \{\ln 1, \ln 2, \ln 3, \ln 4, \dots\}$$

we can find entire functions f_1, f_2, f_3, f_4 such that $Z_{f_1} = A, Z_{f_2} = B, Z_{f_3} = C, Z_{f_4} = D$.

Definition. The Weierstrass elementary factor $E_p(z)$ is defined by $E_0(z) = 1 - z$ for $p \in \mathbf{N}$ ($z \in \mathbf{C}$ and $p = 0, 1, \dots$),

$$E_p(z) = (1 - z) \exp\left(z + \frac{z^2}{2} + \dots + \frac{z^p}{p}\right)$$

If $z_n \neq 0$ and $|z_n| \rightarrow \infty$ then

$$\prod_{n=1}^{\infty} E_n\left(\frac{z}{z_n}\right)$$

is an entire function whose zero set is z_n (at worst, this is an entire function). Notice how $E_p(1) = 0$ and $E_p(0) = 1$.

Lemma. (Weierstrass) If $p \in \mathbf{N}$ and $|z| \leq 1$, then $|E_p(z) - 1| \leq |z|^{p+1}$.

Proof. Since E_p is entire,

$$E_p(z) = 1 + \sum_{n=1}^{\infty} a_n z^n$$

for all z . Thus,

$$E_p'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

By straightforward calculation,

$$\begin{aligned} E_p'(z) &= -\exp\left(z + \frac{z^p}{p}\right) + (1-z) \exp\left(z + \dots + \frac{z^p}{p}\right) \times (1+z+z^2+\dots+z^{p-1}) \\ &= -\exp\left(z + \dots + \frac{z^p}{p}\right) + (1-z^p) \exp\left(z + \dots + \frac{z^p}{p}\right) \\ &= -z^p \exp\left(z + \frac{z^2}{2} + \dots + \frac{z^p}{p}\right) \end{aligned}$$

From this we know that

$$a_1 = a_2 = \dots = a_p = 0$$

and $a_n \leq 0$ for all $n \geq p+1$. Hence, for $|z| \leq 1$,

$$\begin{aligned} |E_p(z) - 1| &= \left| \sum_{n=1}^{\infty} a_n z^n \right| \\ &= \left| \sum_{n=p+1}^{\infty} a_n z^n \right| \\ &= |z|^{p+1} \left| \sum_{n=p+1}^{\infty} a_n z^{n-p-1} \right| \\ &\leq |z|^{p+1} \cdot \sum_{n=p+1}^{\infty} |a_n| \end{aligned}$$

Since $E_p(1) = 0$, we have $1 + \sum_{n=1}^{\infty} a_n = 0$ and thus

$$1 + \sum_{n=p+1}^{\infty} a_n = 0$$

since $a_1 = \dots = a_p = 0$. Thus

$$1 - \sum_{n=p+1}^{\infty} |a_n| = 0$$

since $|a_n| = -a_n$ for $n \geq p+1$. Then,

$$\sum_{n=p+1}^{\infty} |a_n| = 1$$

and

$$|E_p(z) - 1| \leq |z|^{p+1}$$

□

Theorem. (Big theorem) Let $\{z_n\}$ be a sequence in $\mathbf{C}$ such that $z_n \neq 0$ and $|z_n| \rightarrow \infty$. If

$$\sum_{n=1}^{\infty} \frac{1}{|z_n|^{p+1}} < \infty$$

for $p = 0, 1, 2, \dots$ then

$$f(z) = \prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

is an entire function with

$$Z_f = \{z_n \mid n \in \mathbf{N}\}$$

Also,

$$\prod_{n=1}^{\infty} E_n\left(\frac{z}{z_n}\right)$$

is an entire function whose zero set is $\{z_n\}$.

Proof. For every $z \in \bar{D}(0, R)$,

$$\left|E_p\left(\frac{z}{z_n}\right) - 1\right| \leq \left|\frac{z}{z_n}\right|^{p+1} \leq 1$$

for $|z/z_n| \leq 1$. Notice that this is satisfied for all $n \geq K$ since $|z_n| \rightarrow \infty$. And,

$$\sum_{n=k}^{\infty} \left|\frac{z}{z_n}\right|^{p+1} = |z|^{p+1} \sum_{n=k}^{\infty} \frac{1}{|z_n|^{p+1}} \leq M \sum_{n=k}^{\infty} \frac{1}{|z_n|^{p+1}}$$

Thus, by the previous theorem,

$$\prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

is analytic in $D(0, R)$ for every R . Hence,

$$\prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

is entire whose zero set is $\{z_n\}$. Notice how if we put $n - 1$ in p , we prove the second part. □

Example. Using this theorem, show that an entire function f such that $Z_f = \{1, 2, \dots\}$ is

$$f(z) = \prod_{n=1}^{\infty} E_1\left(\frac{z}{z_n}\right) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n}\right) e^{z/n}$$

An entire function g such that $Z_g = \{1, 4, 9, 16, 25, \dots\}$ is

$$g(z) = \prod_{n=1}^{\infty} E_0\left(\frac{z}{n^2}\right) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n^2}\right)$$

Similarly $Z_h = \{1, \sqrt{2}, \sqrt{3}, \dots\}$.

$$h(z) = \prod_{n=1}^{\infty} E_2\left(\frac{z}{\sqrt{n}}\right) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{\sqrt{n}}\right) e^{z/\sqrt{n} + z^2/2n}$$

If $Z_F = \{\ln 1, \ln 2, \ln 3, \dots\}$, there exists no such p , and we must say that

$$F = \prod_{n=2}^{\infty} E_n\left(\frac{z}{\ln n}\right) = z \prod_{n=2}^{\infty} \left(1 - \frac{1}{\ln n}\right) \exp\left(\frac{z}{\ln n} + \frac{1}{2}\left(\frac{z}{\ln n}\right)^2 + \dots + \frac{1}{n}\left(\frac{z}{\ln n}\right)^n\right)$$

LECTURE 22 (DECEMBER 2ND)

Recall. We had $E_0(z) = (1 - z)$ and for $p \in \mathbf{N}$,

$$E_p(z) = (1 - z) \exp\left(z + \frac{z^2}{2} + \dots + \frac{z^p}{p}\right)$$

Theorem. For $|z| \leq 1$,

$$|E_p(z) - 1| \leq |z|^{p+1}$$

Theorem. If $|z_n| \rightarrow \infty$, $z_n \neq 0$ and

$$\sum_{n=1}^{\infty} \frac{1}{|z_n|^{p+1}} < \infty$$

then

$$\prod_{n=1}^{\infty} E_p\left(\frac{z}{z_n}\right)$$

is an entire function whose zero set is $\{z_n \mid n \in \mathbf{N}\}$

Example. Let $A = \{n \mid n \in \mathbf{N}\}$ then

$$f(z) = \prod_{n=1}^{\infty} E_3\left(\frac{z}{n}\right)$$

satisfies $Z_f = A$.

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n}\right) \exp\left(\frac{z}{\sqrt{3}n} + \frac{z^2}{2n} + \dots\right)$$

Example. A function analytic in $U = \{|z| < 1\}$ whose zero set is $\{1 - 1/n \mid n \in \mathbf{N}\}$ is

$$f(z) = g\left(\frac{1}{1-z}\right)$$

Theorem. If D is a bounded domain, $\alpha_n \in D$ with no limit point in D , then there is an analytic function on D with $Z_f = \{\alpha_n\}$.

Proof. For each α_n , there is $\beta_n \in D^c$ such that $|\beta_n - \alpha_n| \leq |\beta - \alpha_n|$ for all $\beta \in D^c$. Suppose $\{\alpha_n\}$ is an infinite set, then $\lim_{n \rightarrow \infty} |\beta_n - \alpha_n| = 0$, otherwise, $\{\alpha_n\}$ has a limit point in D . We'll now show that

$$f(z) = \prod_{n=1}^{\infty} E_n\left(\frac{\alpha_n - \beta_n}{z - \beta_n}\right)$$

is analytic with $Z_f = \{\alpha_n\}$. We now show that

$$f_N(z) = \prod_{n=1}^N E_n\left(\frac{\alpha_n - \beta_n}{z - \beta_n}\right)$$

converges uniformly on every compact subset of D . Let K be a compact subset. Then, there is $\delta > 0$ such that $|z - \beta_n| > \delta$ for all $z \in K$. So, there is M such that

$$\left|\frac{\alpha_n - \beta_n}{z - \beta_n}\right| < \frac{1}{2}$$

for all $z \in K$ and $n > M$. Then, if $n > M$ and $z \in K$, then

$$\left|E_n\left(\frac{\alpha_n - \beta_n}{z - \beta_n}\right) - 1\right| \leq \left|\frac{\alpha_n - \beta_n}{z - \beta_n}\right|^{n+1} \leq \left(\frac{1}{2}\right)^{n+1}$$

Thus,

$$\sum_{n=K}^{\infty} \left|E_n\left(\frac{\alpha_n - \beta_n}{z - \beta_n}\right) - 1\right| \leq \sum_{n=M}^{\infty} E_n\left(\frac{\alpha_n - \beta_n}{z - \beta_n}\right)$$

converges uniformly on K . □

Lemma. We have constantly seen f_n analytic, $|f_n(z) - 1| \leq c_n$ uniformly on a compact set with $\sum c_n < \infty$. Then,

$$f(z) = \prod_{n=1}^{\infty} f_n(z)$$

is analytic. Setting

$$F_N(z) = \prod_{n=1}^N f_n(z)$$

we have

$$\frac{F'_N(z)}{F_N(z)} = \sum_{n=1}^N \frac{f'_n(z)}{f_n(z)}$$

Therefore,

$$\frac{f'(z)}{f(z)} = \sum_{n=1}^{\infty} \frac{f'_n(z)}{f_n(z)}$$

This is due to the fact that $f_n \rightarrow f$ uniformly on K and $f'_n \rightarrow f'$ uniformly on K . Also, f_n is uniformly bounded below on K .

Example. Let

$$g(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right) = f(z^2)$$

where $f(z) = \prod_{n=1}^{\infty} (1 - z/n^2)$ is entire. $Z_g = \{\pm 1, \pm 2, \dots\} = \mathbf{Z} \setminus \{0\}$, all simple zeros. Thus,

$$\sin \pi z = e^{h(z)} z g(z)$$

for some entire function h . Let's apply the logarithmic derivative on this!

$$\sin \pi z = e^{h(z)} z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

Then,

$$\pi \cot \pi z = h'(z) + \frac{1}{z} + \sum_{n=1}^{\infty} \frac{-\frac{2z}{n^2}}{\left(1 - \frac{z^2}{n^2}\right)} = h'(z) + \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}$$

However, we know that

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}$$

and we find $h' = 0$, that is, h is constant. Then, we find h to be π . Now,

$$\frac{\sin \pi z}{\pi z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

Matching the coefficient for z^2 , we get

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi}{6}$$

Definition. (Complex gamma function $\Gamma(z)$) Let

$$f(z) = \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}$$

be an entire function whose zero set is $-\mathbf{N}$. Then, $f(z-1)$ is an entire function whose zero set is $\{0, -1, \dots\}$. Thus, there is an entire function g such that $f(z-1) = e^{g(z)} z f(z)$. Also,

$$\frac{f'(z-1)}{f(z-1)} = g'(z) + \frac{1}{z} + \frac{f'(z)}{f(z)}$$

To find f'/f , we take the logarithmic derivative,

$$\frac{f'(z)}{f(z)} = \sum_{n=1}^{\infty} \frac{\left[\left(1 + \frac{z}{n}\right) e^{-z/n}\right]'}{\left(1 + \frac{z}{n}\right) e^{-z/n}} = \sum_{n=1}^{\infty} \frac{-\frac{z}{n^2}}{\left(1 + \frac{z}{n}\right)} = \sum_{n=1}^{\infty} \frac{-z}{n(n+z)} = \sum_{n=1}^{\infty} \left(\frac{1}{n+z} - \frac{1}{n}\right)$$

where $[(1 + z/n)e^{-z/n}]' = [e^{-z/n} + ze^{-z/n}/n]' = -ze^{-z/n}/n^2$. We now find

$$\sum_{n=1}^{\infty} \left(\frac{1}{n+z-1} - \frac{1}{n}\right) = g'(z) + \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{1}{n+z} - \frac{1}{n}\right)$$

However,

$$\sum_{n=2}^{\infty} \left(\frac{1}{n+z-1} - \frac{1}{n}\right) = \sum_{n=2}^{\infty} \left(\frac{1}{n+z-1} - \frac{1}{n-1} + \frac{1}{n-1} - \frac{1}{n}\right) = \sum_{n=1}^{\infty} \left(\frac{1}{n+z} - \frac{1}{n}\right) + \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$$

and $g'(z) = 0$, and is a constant. Let $g(z) = \gamma$, then

$$f(z-1) = e^{\gamma} z f(z)$$

Putting $z = 1$,

$$1 = f(0) = e^{\gamma} f(1)$$

and

$$1 = e^{\gamma} \prod_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) e^{-1/n} \quad \& \quad e^{\gamma} = \left[\prod_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) e^{-1/n} \right]^{-1}$$

so that

$$\gamma = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{k} - \ln\left(1 + \frac{1}{k}\right)\right) = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \ln(n+1)\right)$$

This constant is called Euler's gamma.

LECTURE 23 (DECEMBER 4TH)

Recall. (Euler's constant) We have defined $f(z) = \prod_{n=1}^{\infty} (1 + z/n)e^{-z/n}$ and $f(z-1) = e^{g(z)}zf(z)$. As $g'(z) = 0$, we set $g(z) = \gamma$ and find:

$$g(z) = \gamma = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln(n+1) \right)$$

Definition. (Weierstrass gamma function) The Weierstrass gamma function is defined as:

$$\Gamma(z) = \frac{1}{e^{\gamma z}zf(z)} = \frac{e^{-\gamma z}}{z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n} \right)^{-1} e^{z/n}$$

where $e^{\gamma z}zf(z)$ is an entire function with simple zeros at $0, -1, -2, \dots$. Therefore, the function has simple poles at these points.

$$\frac{1}{\Gamma(z)} = e^{\gamma z}zf(z) = e^{\gamma z}z \prod_{n=1}^{\infty} \left(1 + \frac{z}{n} \right) e^{-z/n}$$

From $f(z-1) = e^{\gamma z}zf(z)$,

$$f(z) = e^{\gamma}(z+1)f(z+1)$$

so that

$$\Gamma(z+1) = \frac{1}{e^{\gamma(z+1)}(z+1)f(z+1)} = \frac{1}{e^{\gamma z}f(z)} = \frac{z}{e^{\gamma z}zf(z)} = z\Gamma(z)$$

This tells us that $\Gamma(z+1) = z\Gamma(z)$ for all $z \in \mathbf{C} \setminus \{0, -1, -2, \dots\}$. Meanwhile,

$$\Gamma(1) = \frac{1}{e^{\gamma}f(1)} = \frac{1}{e^{\gamma}e^{-\gamma}} = 1$$

and $\Gamma(n+1) = n!$.

Definition. (Euler Gamma function) Now, the Euler Gamma function is defined for $x \in \mathbf{R}$ and $x > 0$ as:

$$\Gamma(x) = \int_0^{\infty} t^{x-1}e^{-t} dt$$

If $x \leq 0$,

$$\int_0^{\infty} t^{x-1}e^{-t} dt \geq \int_0^1 t^{x-1}e^{-t} dt \geq \int_0^1 t^{x-1}e^{-1} dt \geq \int_0^1 \frac{1}{t} \frac{1}{e} dt = \infty$$

Meanwhile, for $z \in \mathbf{C}$ with $\operatorname{Re}(z) > 0$,

$$\Gamma(z) = \int_0^{\infty} t^{z-1}e^{-t} dt$$

and with $z = x + iy$,

$$|t^z| = |e^{z \ln t}| = e^{x \ln t} = t^x$$

tells us that the integral is well-defined. The easiest way to see that the Euler Gamma function is a analytic continuation of the Weierstrass gamma function, we use Euler's identity.

Recall. Recall that

$$\frac{\sin \pi z}{\pi z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right) = f(z)f(-z)$$

so that

$$\sin \pi z = \pi z f(z)f(-z) = \pi z \frac{1}{e^{\gamma z} z \Gamma(z)} \frac{1}{e^{-\gamma z} (-z) \Gamma(-z)} = \frac{\pi}{\Gamma(z)(-z) \Gamma(-z)} = \frac{\pi}{\Gamma(z) \Gamma(1-z)}$$

That is,

$$\Gamma(z) \Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

Theorem. (Euler identity) We see that

$$\Gamma(z) \Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

for $z \in \mathbf{Z}$.

Proof. For $0 < s < 1$,

$$\Gamma(s) = \int_0^{\infty} t^{s-1} e^{-t} dt \quad \& \quad \Gamma(1-s) = \int_0^{\infty} u^{-s} e^{-u} du$$

Putting $u = ty$ for $t > 0$,

$$\Gamma(1-s) = t \int_0^{\infty} (ty)^{-s} e^{-ty} dy$$

Consider

$$\int_0^{\infty} t^{s-1} e^{-t} \left[\int_0^{\infty} t (ty)^{-s} e^{-ty} dy \right] dt = \Gamma(s) \Gamma(1-s)$$

which is also equal to

$$\int_0^{\infty} \int_0^{\infty} e^{-(1+y)t} y^{-s} dt dy = \int_0^{\infty} \frac{y^{-s}}{1+y} dy = \frac{\pi}{\sin[(1-s)\pi]} = \frac{\pi}{\sin \pi s}$$

Therefore,

$$\Gamma(s) \Gamma(1-s) = \frac{\pi}{\sin \pi s}$$

At this stage, using the identity theorem along the real line,

$$\Gamma(z) \Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

for $0 < \operatorname{Re}(z) < 1$. □

Definition. (Euler zeta function) The Euler zeta function is defined as:

$$\xi(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

for $s > 1$ if real or $\operatorname{Re}(s) < 1$ if complex. Consider $\operatorname{Re}(s) = \sigma > 1$.

$$\begin{aligned} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx &= \int_0^{\infty} x^{s-1} \sum_{n=1}^{\infty} e^{-nx} dx = \sum_{n=1}^{\infty} \int_0^{\infty} x^{s-1} e^{-nx} dx \\ &= \sum_{n=1}^{\infty} \int_0^{\infty} x^{s-1} e^{-nx} dx = \sum_{n=1}^{\infty} \frac{1}{n^s} \int_0^{\infty} y^{s-1} e^{-y} dy \\ &= \xi(s) \Gamma(s) \end{aligned}$$

by setting $nx = y$. Thus we have the important identity that

$$\xi(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx$$

Definition. (Dirichlet eta function) The function is defined as:

$$\eta(s) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^s} = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \dots$$

which converges in $\operatorname{Re}(s) > 0$. This is closely related to $\xi(s)$ as

$$\xi(s) - \eta(s) = 2 \sum_{n=1}^{\infty} \frac{1}{(2n)^s} = 2^{1-s} \xi(s)$$

and

$$\xi(s) = (1 - 2^{1-s})^{-1} \eta(s)$$

when $\operatorname{Re}(s) > 1$. Notice that this version of the zeta function has no roots in $0 < \operatorname{Re}(s) < 1$ unless

$$s = \frac{1}{2} + it$$

Lemma. (Prime number theorem) The prime number theorem states that:

$$\lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1$$